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HAND POSE PREDICTION DRIVEN BY MOTONEURON SYNERGIES

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Chapter 1

Introduction

1.1 Problem Description

Stroke is a leading cause of disability worldwide, with a substantial personal, social, and economic impact. A frequent consequence is hemiplegia, characterized by weakness or paralysis of the upper and lower limbs on one side of the body, typically contralateral to the cerebral lesion, which severely impairs upper limb function and manual dexterity (Park et al. 2025 [1]).

Upper limb motor impairment arises not only from direct damage to descending motor pathways, such as the corticospinal tract conveying voluntary motor commands from the cortex to spinal motoneurons, but also from neurophysiological alterations in cortical dynamics that affect the generation, organization, and selectivity of motor commands (Lang et al., 2003 [2]).

A key mechanism underlying these alterations is the disruption of interhemispheric balance (Park et al. 2025 [1]). Under physiological conditions, the two motor cortices exert a balanced reciprocal inhibitory control. After stroke, this balance is often lost: the lesioned hemisphere exhibits reduced excitability, while the contralesional (intact) hemisphere may exert excessive inhibitory influence over the affected motor cortex (Murase et al. 2025 [3]). This abnormal interhemispheric inhibition further suppresses motor output from the damaged hemisphere, limiting voluntary activation of distal muscles.

Together, impaired command transmission and altered cortical motor control contribute to muscle weakness, reduced coordination, and loss of dexterity, making functional recovery of the upper limb particularly challenging.

Restoring upper limb motor function therefore requires therapeutic interventions capable of promoting neuroplastic reorganization, rebalancing interhemispheric activity, and establishing a functional coupling between motor intention and movement execution.

Despite the availability of conventional rehabilitation approaches, many patients continue to experience persistent deficits in hand function, grasp control, and fine motor skills. Traditional rehabilitation methods, although effective, are limited by factors such as therapist or patient fatigue and time constraints, making it difficult

to maintain the precise levels of intensity and repetition required for optimal recovery. These limitations affect independence in activities of daily living (ADLs) and highlight the need for more effective, individualized, and responsive rehabilitation strategies (Park et al. 2025 [1]).

1.2 Background and Motivation

To properly evaluate the impact of rehabilitation strategies, it is essential to first define the temporal progression of recovery. According to the consensus established by the Stroke Recovery and Rehabilitation Roundtable (Bernhardt et al. [4]), the post stroke timeline is categorized into distinct phases: the acute phase (within a week of stroke onset), the subacute phase (from 7 days to 6 months), and the chronic phase (beyond 6 months). The effectiveness of advanced interventions, such as Robot Assisted Therapy (RAT) and immersive virtual environments, has been extensively studied in relation to these time windows. Recent meta-analyses confirm that while motor benefits derived from RAT are consistent across all recovery phases, the magnitude of the effect tends to be greater in the acute and early subacute stages across both proximal (shoulder and elbow) and distal (wrist and hand, including the fingers) segments of the upper limb (Park et al. 2025 [1]). This heightened efficacy is consistent with the theory that neuroplasticity is most pronounced in the initial post-stroke period, particularly within the first 30 days. Despite these temporal differences, these advanced technologies remain crucial tools throughout the continuum of care.

Although robot assisted therapy consistently produces statistically significant improvements in upper limb motor function, the magnitude of these effects is generally small. This highlights the need for complementary strategies capable of enhancing functional engagement for clinically meaningful improvements.

The observed benefits of robot assisted therapy are generally attributed to the combined effect of intensive, repetitive, task oriented movements. In addition, as illustrated in Fig. 1.1, continuous multimodal sensorimotor feedback reinforces motor intention and supports neuroplastic adaptation from experience: for example virtual reality and mirror therapy or auditory or haptic feedbacks.
However, most conventional robotic systems rely on externally imposed or pre programmed movement assistance, which may limit the degree of active neural engagement. In this context, approaches that directly harness the patient's residual voluntary neural drive may further enhance cortical activation and motor relearning.

Building on these principles, systems powered by surface electromyography (sEMG) are emerging as effective non invasive bioelectrical solutions for immersive rehabilitation and advanced robotic applications, including prosthetic devices and robotic orthoses (Rozevink et al. 2024 [5]). sEMG is inherently suited for human computer interface (HCI) applications, as it captures the anticipatory muscle activation that precedes movement execution. This enables intention driven control, reduces interaction latency, and promotes a tighter coupling between motor com-

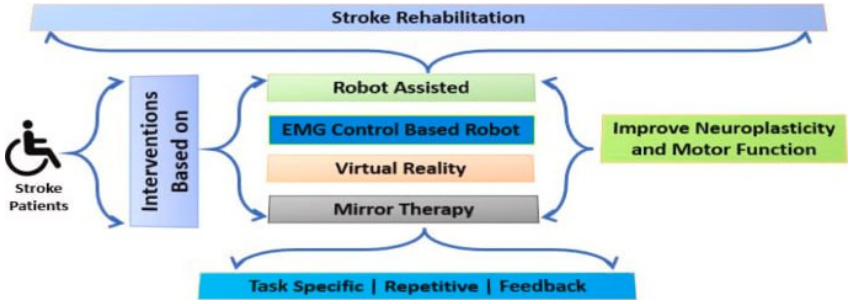


Figure 1.1 Combination of different kind of interventions improve stroke rehabilitation (Amin et al. 2024 [9]).

mand generation and sensory feedback (Tacca et al. 2024 [6]).

sEMG provides direct access to the user's motor intention by capturing muscle activation patterns even in individuals with severely compromised movement.
This is particularly valuable for the primary target groups of this work: (i) post-stroke individuals in the acute and subacute phases, who often exhibit reduced hand function and grip strength but retain residual voluntary activation detectable via sEMG, and (ii) older adults or individuals with hand dysfunction due to orthopedic injuries or age related decline who benefit from focused strengthening and retraining of fine motor skills.

The sEMG driven approach also holds significant potential for individuals with upper limb amputations. In transradial (below the elbow) amputations, the preservation of forearm musculature provides accessible and functionally specific myoelectric signals. Studies such as the one documented by Mereu et al. [7] demonstrate that voluntary phantom limb movements impose behavioral constraints similar to real movements, even many years after amputation. This finding suggests that residual neuromuscular activity can still encode meaningful motor intent and serve as a reliable source of control signals.
However, sEMG alone may not always provide sufficient information for effective control in higher level amputations such as transhumeral (above the elbow) or shoulder disarticulation; in such cases, advanced techniques such as Targeted Muscle Reinnervation (TMR), Regenerative Peripheral Nerve Interface (RPNI) along with neuromusculoskeletal interfaces, wireless Implantable MyoElectric Sensor (IMES) can be employed to create new, accessible control sites (D'Accolti et al. 2023 [8]).

In all these scenarios, the central challenge is not only the acquisition of myoelectric signals, but their transformation into meaningful motor outputs. In order to control a prosthetic device or to provide effective feedback in rehabilitation, the recorded sEMG signals must be mapped into kinematic variables (e.g., joint angles, hand posture) or kinetic variables (e.g., force). This mapping step is therefore essential, as it defines how neural intent is translated into observable or controllable movement.

Mapping raw sEMG signals to complex hand kinematics presents significant computational challenges due to the stochastic nature of the signal and the nonlinear relationship between muscle activation and hand posture or force generation. Traditional approaches have largely focused on decoding movement intention into discrete movement categories (e.g., “open hand” versus “close hand”). While often robust, such strategies constrain the user to a predefined set of gestures and do not provide the continuous, proportional control required for natural manipulation and effective neurofeedback. In response to these constraints, several approaches have been proposed including regression models that map sEMG features to kinematic or force outputs. More recently, deep learning techniques have been increasingly adopted to model the complex nonlinear relationships between muscle activation and movement, enabling the prediction of fluid hand trajectories and grasp forces. Despite methodological differences, the vast majority of these approaches rely on global surface EMG features computed in the time, frequency, or time–frequency domains.

The present work is based on the hypothesis that shifting the input domain from conventional global sEMG features to neural drive related features derived from sEMG decomposition and the output domain from individual joint angles to coordinated kinematic synergies, may enable the development of a framework that is not only more accurate, but also more consistent with the physiological organization of human motor control.

1.3 Objectives

This research investigates the feasibility of decoding forearm neuromuscular activity from high density surface electromyography HD sEMG to continuously predict hand grasp poses. Specifically, the thesis pursues the following computational and experimental goals:

- to analyze an open source HD sEMG and kinematic dataset, adapting motoneuron decomposition protocols for quasi isometric contractions in order to extract reliable and consistent motor unit spike trains;
- to apply dimensionality reduction techniques to the extracted motor unit activations, identifying motoneural synergies that provide a robust and low dimensional input space;
- to design, train, and evaluate a recurrent deep learning architecture capable of continuously mapping the extracted neural synergies to multi channel hand kinematics across single task and concatenated task scenarios;
- to implement a kinematic retargeting pipeline within a Unity environment, converting the network predicted glove sensor channels into local bone rotations for the visual validation of the decoded hand poses on a 3D rigged virtual model.

Together, these objectives aim to establish a comprehensive methodological framework for future developments in immersive rehabilitation, soft robotic hand therapy, and advanced myoelectric assistive control systems.

1.4 Outline

The thesis is organized as follows.

Chapter 2 introduces the physiological foundations of the surface EMG signal and provides a comprehensive review of the State of the Art in sEMG driven hand grasp prediction. It outlines the evolution of control strategies, rom early direct proportional control methods, to Pattern Recognition techniques for movement classification through Machine Learning algorithms, and up to recent robust Deep Learning classification or continuous regression approaches. A core focus of the chapter is dedicated to the concept of neural synergies, which provide a low dimensional representation of motor control. The chapter highlights how decoding motor unit activity from HD sEMG signals to extract motoneural synergies simplifies the input space for neural networks, enabling a more natural and intuitive mapping for hand pose prediction. Finally, it briefly reviews previous works attempting to map neural information to hand kinematics and discusses the role of sensory feedback in immersive rehabilitation.

Chapter 3 details the experimental framework and the computational pipeline developed to map the neural drive into continuous kinematic control. After a brief overview of the preliminary activities carried out during the internship at Fondazione Santa Lucia (IRCCS), where the focus was on establishing an experimental protocol for the simultaneous acquisition of HD sEMG and hand kinematic data, the chapter introduces the open source dataset used in the experiment. The methodology is then structured into three main phases: the motoneuron decomposition to extract individual firing patterns for isometric contractions, the dimensionality reduction to identify the neural synergies forming the control manifold, and the design of the Long Short Term Memory (LSTM) model for temporal pose prediction. Furthermore, the chapter describes the integration of the system within a virtual reality environment, explaining the spatial mapping required to convert the predicted glove sensor channels into local bone rotations for a 3D rigged hand model in Unity.

Chapter 4 presents the experimental results and evaluates the performance of the proposed architecture. It analyzes the extracted motoneuron synergies and details the predictive capabilities of the LSTM model during both single task and concatenated task scenarios, focusing on the reconstructed isometric contractions. The results are discussed across both intra subject and cross subject evaluation paradigms. Finally, the chapter concludes by discussing the methodological limitations encountered in the experiments and outlines potential future developments for dynamic tracking and real time applications.

Chapter 2

Theoretical Starting Point

2.1 sEMG principles

The neurophysiological principles underlying sEMG are well defined in the literature and have been extensively described in studies on neuromuscular control. Within the domain of neuroprosthetic interfacing, these concepts are clearly summarized by Capsi-Morales et al. (2025) [10] in a real time study involving a 2-DOF prosthetic hand controlled by two postural synergies, providing a coherent framework for understanding how sEMG reflects neural drive to the muscles.

Muscle contraction does not originate directly in the muscle tissue itself, but arises from the activity of motor units (MUs), the fundamental functional elements of the neuromuscular system. A motor unit consists of a single α-motor neuron, whose cell body resides in the spinal cord, and all the muscle fibers innervated by its axon [1]. Each motor unit innervates only a portion of a muscle, and so motor unit architecture varies substantially across them.
In particular, hand function relies on the coordination between extrinsic and intrinsic muscles. Intrinsic muscles, responsible for dexterity and fine motor control, such as those of the hand, are characterized by small motor units innervating fewer than ten fibers per neuron, enabling precise force modulation. In contrast, extrinsic muscles, located in the forearm and involved in gross force production, rely on large motor units that may innervate hundreds or even thousands of fibers (Heckman and Enoka, 2012 [12]).

Voluntary movement is therefore controlled by the central nervous system (CNS), through descending commands originating in cortical and subcortical areas and transmitted via spinal motor neurons.
Each discharge of an α-motor neuron produces a motor unit action potential (MUAP). At the muscle surface, MUAPs appear as spatiotemporal electrical patterns whose amplitude and waveform depend on the number, size, and spatial distribution of the activated muscle fibers relative to the recording electrodes.
The spatial resolution of these recordings plays a crucial role in the amount of

[1] A portion of a nerve cell that carries nerve impulses with other neurons or with muscle or gland cells [11].

information that can be extracted. As reported by Li et al.(2021) [13], this leads to a fundamental distinction in recording setups:

- Sparse sEMG systems (2–16 channels): these take an accurate anatomic localization strategy over the muscle. They have less data to transfer and low cost of hardware resources; however, they are very sensitive to the domain change of sEMG signal, such as electrode displacement or variations in limb posture. But the use of fewer electrodes would allow for easier physical integration inside robotic assistive or prosthetic socket (Li et al., 2023, [14]).
- High-density sEMG (HD-sEMG, 32–256 channels): These systems employ dense electrode arrays, positioned on a single or only a few muscles (Tacca et al., 2024, [6]), enabling detailed mapping of MUAP propagation. This high resolution is less susceptible to performance issues due to variation in electrode placement and becomes particularly informative, as it reflects the anatomical region of activation and supports advanced decoding and control strategies. The signal is recorded between a single active electrode and a common reference electrode (monopolar configuration). However, the escalation in electrode density necessitates more sophisticated analog front-end designs and substantial computational overhead for signal processing.

The surface EMG signal is interpreted as the linear superposition of MUAP trains generated by individual motor units, in addition to measurement noise (Chen et al., 2020 [15]). By leveraging the information provided by high-density recordings, each MUAP train can be modeled as the temporal convolution of two components: a spike train representing the motor-unit firing activity (MUST) and the characteristic MUAP waveform, which reflects the electrophysiological response of the innervated muscle fibers (Costa et al., 2025, [16]). This interaction, in time domain, is formalized by the Convolutive EMG Mixing Model (Fig.2.1), which represents a Multiple-Input Multiple-Output (MIMO) system where the anatomical properties are embedded in the convolution kernels, the MUAPs. That model is mathematically expressed by the equation (2.1):

$$sEMG_j(t) = \sum_{i=1}^{n} (MUST_i(t) * MUAP_{i,j}(t)) + e(t) \tag{2.1}$$

$sEMG_j(t)$ is the signal recorded by the j-th electrode, $MUST_i(t)$ represents the spike train of the i-th motor unit, $MUAP_{i,j}(t)$ is the spatiotemporal potential of that unit as seen by the electrode, and $e(t)$ denotes the additive measurement noise. From the identified MUSTs, it is possible to extract the Discharge Rate (DR), also referred to as the Firing Rate (FR) of each motor unit. The discharge rate $DR_i(t)$ for the i-th motor neuron is derived from the Inter-Spike Interval (ISI), which represents the time elapsed between two consecutive discharges, as formalized in equation (2.2):

$$DR_i(t) = \frac{1}{ISI} = \frac{1}{t_{i,j} - t_{i,j-1}} \tag{2.2}$$

Instead, Smoothed Discharge Rates represent a continuous approximation of the instantaneous firing rate of the i-th motor neuron. To estimate the instantaneous

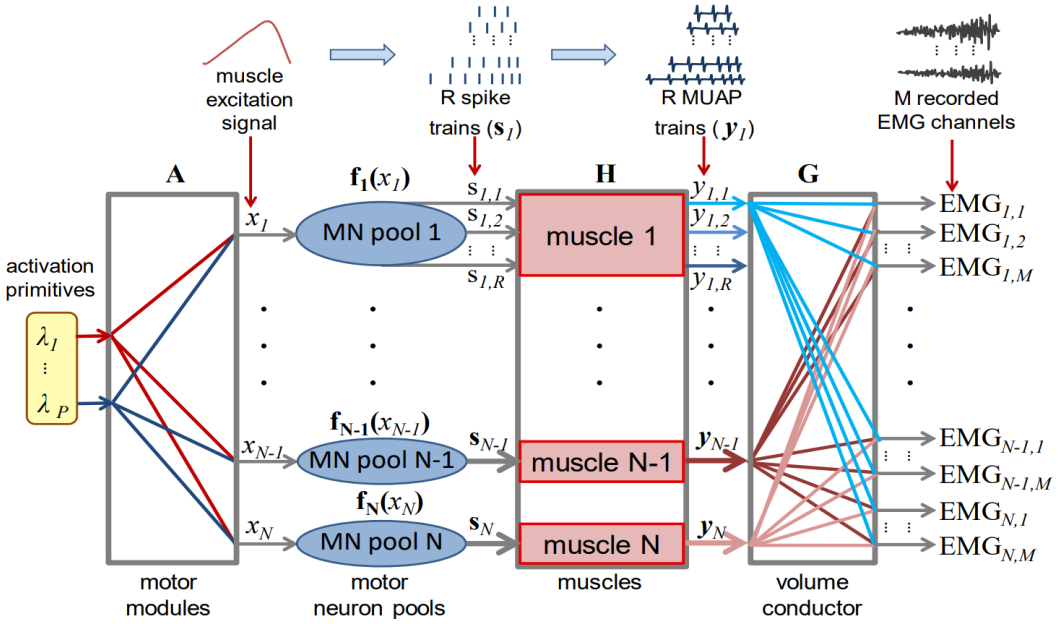


Figure 2.1 Convolutive EMG mixing model (Holobar et al., 2014 [17])

firing rate, each binary sequence of the spike trains is low passed filtered with a fourth order Butterworth filter with cutoff frequency $f_c = 2.5$ Hz. This cutoff frequency corresponds approximately to the bandwidth obtained using a 400-ms Hanning window (Capsi-Morales et al.), a standard duration used to extract the common neural drive from motor unit spike trains.
To ensure comparability across motor units, each SDR is normalized between 0 and 1:

$$\widetilde{\mathrm{SDR}}_i(t) = \frac{\mathrm{SDR}_i(t) - \min(\mathrm{SDR}_i)}{\max(\mathrm{SDR}_i) - \min(\mathrm{SDR}_i)}.$$

2.1.1 Pre-Processing raw sEMG Signal

The recorded sEMG typically exhibits peak-to-peak amplitudes on the order of 0–10 mV, corresponding to approximately 0–1.5 mV in terms of root mean square (RMS) value, with dominant energy in the 20–450 Hz band (Parajuli at al., 2019, [18]). Raw EMG recordings are inherently noisy and highly susceptible to various sources of interference, so, in order to enhance signal quality before feature extraction, pre processing is a crucial stage.
First, notch filtering is often employed to mitigate power line interference, narrow-band artifact at 50 Hz in European systems or 60 Hz in North American systems, along with their harmonics, and, in case of an adaptive filter, it can also remove other frequencies with abnormal peak.
Then, band pass filtering is used to isolate the frequency content of interest in sEMG signals and to suppress low frequency components related to electrode movement and body motion, as well as high frequency noise introduced by the acquisition

system.

Beyond these standard filtering procedures, the subsequent pre-processing stages are strictly dependent on the specific information needed to be extracted from the sEMG signal. While global amplitude features might require further smoothing or rectification to obtain the signal envelope, preserving the raw signal's integrity is essential for decomposition algorithms to identify the underlying neural drive.

2.1.2 Steady and Transient sEMG State

The temporal characteristics of the sEMG signal depend strongly on the phase of muscle activation as also reported by D'Accolti et al. (2023) [8] in decoding human motor intentions by processing electrophysiological signals. The author clearly divides the steady and transient state of the sEMG signal.

During steady-state contraction, the muscle maintains a relatively constant force level over time. From a neural standpoint, this apparent stability does not correspond to a fixed activation pattern. Instead, the CNS continuously adjusts motor unit recruitment and firing rates to compensate for fatigue, mechanical fluctuations, and force stabilization requirements. As a consequence, steady-state sEMG exhibits weak temporal structure and high variability, often resembling a quasi-stationary or stochastic signal. Despite this limitation, steady-state sEMG remains useful for applications such as continuous control and force estimation, where the amplitude of the EMG signal is approximately proportional to the generated force (Mereu et al, 2023, [7]).

In contrast, the transient phase of muscle activation corresponds to the short time interval during which the muscle transitions from rest to contraction (onset) or from contraction back to rest (offset) (Tacca et al., 2024, [6]). During this phase, the rapid recruitment of motor units and abrupt changes in firing rates introduce rich and highly structured temporal dynamics into the sEMG signal. These transient patterns are strongly linked to movement intention and are often more discriminative of specific actions than steady-state activity. For this reason, transient sEMG has gained increasing attention for low-latency and anticipatory control.

The accurate detection of movement onset is a fundamental prerequisite for sEMG processing. As reviewed in the literature, Onset Detection Algorithms (ODAs) vary significantly in how activation thresholds are defined. Common strategies include referencing the signal magnitude at rest, using a percentage of the Maximum Voluntary Contraction (MVC), employing statistical models or the analysis of kinematic trajectories through data gloves (Mereu et al.) [7]. Among these, the double-threshold method remains a standard for its ability to filter transients by requiring the signal to remain above a baseline for a predefined time window.
However, to enhance sensitivity toward incipient muscular contractions, D'Accolti et al. (2022, [19]) adopts a more sophisticated ODA based on an individually calibrated threshold detector that operates on the derivative of the average rectified value. Unlike simple amplitude-based methods, analyzing the derivative allows for the

detection of the initial "surge" of neural activity, effectively capturing the transition phase with higher temporal precision.

To sum up, the physiological and technical foundations of sEMG provide a rich substrate for decoding hand motion, but also present significant challenges that must be addressed through careful signal processing, robust feature extraction, and advanced modeling techniques.

2.2 sEMG to hand pose mapping

Seminal works by D'Accolti et al. [19], Li et al. [13] and Lunardini et al. [20] provide an overview of the early approaches of mapping sEMG signals to hand degrees of freedom, highlighting the evolution from direct control and manual pattern recognition to deep learning methods.

2.2.1 Direct Control

Because of the anatomical preservation of the forearm muscles responsible for hand and wrist motion, primarily flexors and extensors, in individuals with significant impairment or limb loss, sEMG became the foundation of the earliest and still most widespread control strategy for transradial prostheses: direct control.
In its classic two-state formulation, originating from the work of Bottomley et al. (1960) [21], direct control uses a pair of agonist/antagonist muscles to proportionally modulate a single degree of freedom (typically hand open/close) of a motorized prosthetic hand. In this framework, the amplitude of the sEMG signal extracted from each muscle directly determines the control input: higher muscle activation (and thus higher EMG amplitude) produces higher actuation velocity or force in the corresponding direction. The user increases activation of one muscle to close the prosthetic hand and activates the opposing muscle to open it.
Variants of this approach include simple threshold-based and onset detection strategies (Parajuli et al., 2019 , [18]). Onset analysis, as already explained, detects the initiation of muscular activity, identified as a rapid increase in EMG amplitude, enabling simple binary triggers. Similarly, threshold control issues a command only when the EMG signal surpasses a predefined activation boundary.

While intuitive and robust, this strategy is intrinsically limited: it can only discriminate between two opposite actions and neglects simultaneous control when extended to a multi-DoF or multi-grasp system in which multiple muscles act on each DOF (D'Accolti et al., 2022, [19]). Managing more complex tasks thus requires sequential switching between functions, performing a co-contraction (Li et al., 2023,[14]), which increases cognitive load and reduces usability leading to the rejection of the device itself (Parajuli et al., 2019 , [18]).

2.2.2 Pattern Recognition

The limitations of direct control motivated the development of myoelectric pattern recognition (PR) approaches as first proposed by Finley et al. (1967) [22] which rely on the hypothesis that different intended movements correspond to distinguishable and repeatable multi-channel EMG patterns. By extracting features from an array of EMG signals recorded during steady contraction, these systems enable classification of multiple grasp types or wrist motions providing independent control of multiple DOFs using simultaneous, sequential or semi-sequential control, an important step toward more natural control (Parajuli et al., 2019, [18]).

sEMG-based PR systems can be implemented through two broad categories of techniques: (i) traditional machine learning (ML) for feature engineering, and (ii) deep learning (DL) for automatic feature learning.
Regardless of whether ML or DL is used, PR-based myoelectric control generally follows a common processing structure:

1. Data segmentation (windowing): the preprocessed sEMG is divided into short temporal windows through two different types of techniques: non-overlapping (adjacent) window and overlapping window (Parajuli et al.,2019, [18]).
   In the latter, each window overlaps with the previous one by a set increment, which determines the system's update rate. It creates the discrete data blocks necessary for extracting features or for time-series modeling.
   The selection of the window length is a critical design trade-off. As noted by Li et al., while longer sequences provide more robust statistical information and higher classification accuracy, they also introduce a significant processing delay. For a prosthetic system to feel natural and reliable, the total delay, from signal generation to the execution of the command, must not exceed the 300 ms threshold of human perception (Hudgins et al., 1993 [23]). Specifically, Englehart and Hudgins (2003) [24] suggest that window lengths between 150 and 250 ms offer the best balance for mechanical actuators, whereas Nielsen et al. (2011) [25] demonstrated that performance significantly declines with windows shorter than 100 ms due to insufficient data.
   In this context, the overlapping windowing technique is often preferred for high-performance hardware. By carefully setting the window increments (e.g., 50 ms as in the work of D'Accolti et al. [8]), the system can generate a dense and rapid flow of decisions, accelerating the controller's response without sacrificing the length of the analysis window, allowing for the so called continuous classification (D'Accolti et al., 2023, [8]).

2. Feature extraction: in ML-based PR, handcrafted features are computed from each window after necessary signal pre-processing. These features can be extracted in the time domain (e.g., mean absolute value, waveform length), frequency domain (e.g., median frequency, mean frequency), or time-frequency domain (e.g., wavelet transform coefficients) (Li et al.). Typically 'shallow' ML models heavily depends on this selection, as they often struggle to generalize within complex feature spaces without manual intervention (Putro et al., 2024,

[26]).
This complexity increases further when considering that a feature might be effective on one sensor but noise-sensitive on another, especially under varying conditions.
Furthermore, the effectiveness of these feature-channel pairs (the specific combination of a feature extracted from a particular electrode) is strictly tied to their robustness against factors such as limb position variation. To navigate this high-dimensional space, researchers have historically relied on automated selection filters to identify which specific combinations remain stable across different arm orientations (Al-Angari et al., 2016), [27]), but identifying these optimal subsets requires significant computational effort and domain expertise.

To address these issues, DL-PR was introduced, which integrates feature extraction directly into the network architecture, learning hierarchical representations from raw, or minimally processed, sEMG (Li et al.). A particularly innovative approach involves representing sEMG data as images to leverage the spatiotemporal characteristics inherent in HD-sEMG (Atzori et al.)

3. Feature classification: the extracted features or learned representations are fed into a classifier to predict the intended movement class for both hand gesture or grasping.
Typical ML classifiers include Linear discriminant analysis (LDA), K-Nearest Neighbors (KNN), Support vector machine (SVM) (Parajuli et al., 2029, [18]), Random Forests (Atzoti et al., 2016, [28]). While ML remains computationally lightweight and widely adopted in real-time systems, its reliance on engineered features limits its ability to capture the full complexity and variability of sEMG.

   With DL, the same network that performs feature extraction also performs prediction: the high-level representation produced by the backbone is passed directly to the final fully connected layers that output the pose.
   Atzori et al. provided one of the first large-scale evaluations of Convolutional Neural Networks for myoelectric control with NinaPro (Non-Invasive Collaborative Interface for Prosthesis Control) database [29] which provides synchronized recordings of muscular activity, from a maximum of 16 electrodes, hand pose labels and hand kinematics (acquired with CyberGlove II), and force from both intact subjects and transradial amputees. They tested different architectures on 50 different movements in the database. First, it was demonstrated that even a simple CNN (based on LeNet), automatically learning spacial features, could achieve results comparable to traditional machine learning methods like SVM and Random Forest. However, because sEMG signal is a temporal series, hybrid architecture such CNN-RNNs allow to extract both spatial and temporal feature. Also Temporal Convolutional Networks (TCN) have been trained to captured long-term dependencies in sEMG signals. Unlike RNNs, TCNs utilize one-dimensional convolutional layers that run along the time dimension, allowing the model to learn temporal hierarchies more efficiently. Instead, to reduce manual data labeling, Unsupervised Learning approaches

were studied: Autoencoders (AE) could recover the original data as lossless as possible from the abstract data.

To overcome the intrinsic limitations of individual sensing modalities, recent research has shifted toward Multimodal Fusion. For example, sEMG and Computer Vision systems alone offer contextual information, but are affected by different factors, the latter in particular is susceptible to occlusions, varying lighting conditions, and "visual noise." To address this, Zandigohar et al. (2024) [30] proposed a robust framework based on Bayesian evidence fusion, designed to infer grasping intent by synchronizing physiological and environmental data. Their visual branch relies on a CNN based on the YOLOv4 architecture, fine-tuned for grasp detection, to infer grasp-type probabilities from scene images and user gaze. In parallel, the EMG branch processes dynamic sEMG signals by first segmenting the data into movement phases (reaching, grasping, returning, resting) using the Greedy Gaussian Segmentation, and then classifying grasp intentions using an ensemble Extra-Trees classifier trained on time-domain features. The probabilistic outputs of the two modalities are combined through a Bayesian maximum-likelihood formulation, enabling robust and anticipatory grasp intent prediction, particularly during the reaching phase, where early decision-making is critical for hand control. Results demonstrated that fusion is superior to the two modalities alone in each of the movement phases.

4. Post-processing: to remove errors and misclassifications some techniques as Majority Voting (MV), selecting the most frequent choice, and decision based velocity ramp, to attenuate joint speed and ensure a smoother transition between motion states, are employed (Parajuli et al. 2019 [18]).

Although continuous PR is considered the state of the art in clinical research, its applicability remains challenging because the repeatability of the steady-state EMG is statistically weak due to several physiological factors (D'Accolti et al., 2023, [8]). In fact, maintaining a "constant" contraction does not correspond to a constant neural drive as already highlighted in 2.1.2. As a result, similar contractions may produce noticeably different steady-state EMG patterns.
Instead, different studies proposed using the transient state for predicting and control a multi-grasp system with better real-time performance. D'Accolti et al. (2023) [8] demonstrated that the onset of contraction includes relevant information predictive of the pre-planned grip force. They reached a SVM classification decision for hand pose classes in 200 ms after the onset for which the command is maintained active allowing the user to focus only on the following proportional control phase where the velocity of the movement is modulated by the EMG intensity.

2.2.3 Domain Shift

The sEMG control is strictly tied to its robustness across different conditions, a challenge often categorized into three evaluation scenarios (Li et al. [13]) (see Figure 2.2).

1. Intra-session data, used for both training and testing a model, are collected within the same session of different trial for the same subject without removing

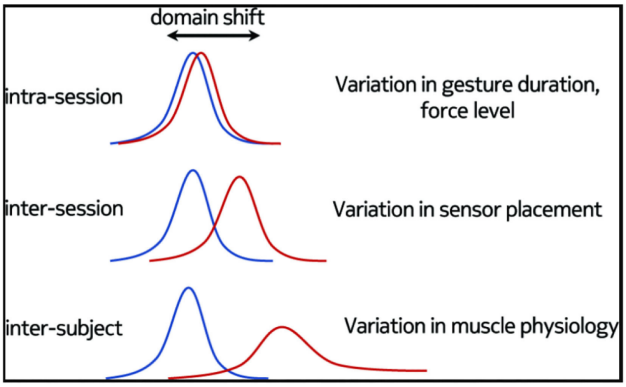


Figure 2.2 Causes of variability of data distribution (Li et al., 2021, [13], Ketyko et al., 2019, [32])

the electrodes, because the signal can change for muscle fatigue, sweat and electrode heating. Robustness is assessed by training on early blocks and testing on the final, fatigued blocks.

2. Inter-session data involve, for the same subject, the reattachment of electrodes and physiological changes between days in order to introduce significant signal variability in the training of the model, for electrode repositioning, baseline shift and skin changes. The model is trained on data from a first session and tested on the successive ones.

3. Inter-subject data are the most critical challenge. Models trained on one participant and tested on another one show substantially worse performance due to a major domain shift (Kaifosh et al., 2025, [31]). This shift is driven by differences in subcutaneous fat thickness, muscle anatomy, neural recruitment, and electrode placement relative to the motor points. Increasing training data for a specific individual substantially improves offline performance, but the benefit is only marginal when attempting to transfer knowledge across different people. Consequently, modern research is shifting towards transfer learning to bridge this gap without requiring exhaustive retraining for every new user. Another approach of transfer learning is fine-tuning a model pre-trained on a vast multi-user dataset with a small amount of labeled data from a specific target user.

4. Force Robustness is considered by performing tasks at varying MVC levels (e.g., low, medium, high force).

2.2.4 Regression Control

A fundamental evolution in HMIs is the transition from discrete gesture classification to continuous movement regression. Focusing on identifying a limited set of predefined gestures (e.g., "fist," "pinch," "point") which as noted by Koch et al., limits the user's expressive range, results in unintuitive control, and fails to replicate the vast motion spectrum of a natural human hand.

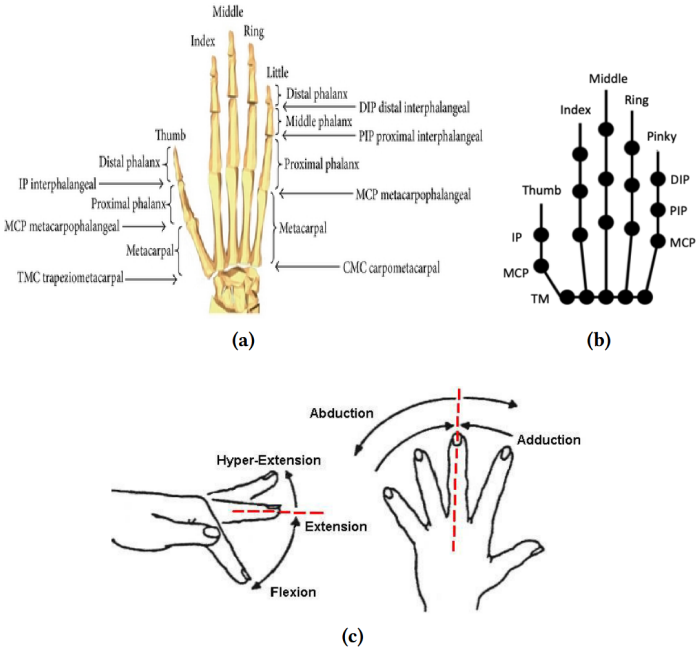


Figure 2.3 Hand anatomy (Li et al.).

To address these limitations, research has investigated whether deep learning architectures, originally successful in classification, can achieve better results to constantly predict the movements of all DoF in the hand, 24 DOF, (Fig. 2.3) and wrist, 2 DoF, to account for flexion/extension and abduction/adduction.

For example, Koch et al. (2020) [33] compared three RNN variants for this task with NinaPro's datasets: a single Long Short-Term Memory (LSTM), stacked LSTMs, and a ConvLSTM. They found that a single LSTM cell often outperformed more intricate architectures, but this may be attributed to the relatively limited scale of the datasets used, where the risk of overfitting in complex networks like ConvLSTMs is higher.

Systems like Neuropose (Li et al., 2021, [13]) tackle the regression problem when more practile, but simpler, wearable sEMG acquisition device are employed, such Myo Armband that captures the raw sEMG signals through 8-channels and a 9-axis IMU. The signals, acquired for 11 participants that performed 15 minutes of different hand movements each, are given as input to a modified U-Net architecture strengthen by residual bottleneck layers. While U-Nets are famous in image segmentation, here they serve as a powerful "spatiotemporal translator."
To train the model, researchers synchronized the sEMG data with ground truth hand poses captured by a depth camera. This alignment is managed through Coordinated Universal Time (UTC) timestamps, ensuring that every electrical burst from the muscles is precisely mapped to the corresponding physical posture in a 3D space.
The network processes 5-second windows of sensor data. First, the convolutional encoder, a series of CONV-BN-ReLU-MaxPool layers, compresses the raw EMG

stream and extract deep, hierarchical features while filtering out signal noise. These features are then refined through residual blocks, which allow the network to learn complex, non-linear mappings, through skip connections, without losing the signal's integrity, avoiding the vanishing gradient problem. Finally, a decoder reconstructs the hand pose at the original sampling frequency. But, While reconstructing the complex sequence of 21 finger joint angles, the model specifically predicts 16 dimensions directly, deriving the remaining angles through known biomechanical constraints to ensure a realistic pose.
To achieve its high level of precision, NeuroPose employs a total loss computed as the sum of individual errors across the hand's joints:

$$Loss = Loss_{MCP_f/e} + Loss_{MCP_aa} + Loss_{PIP} + Loss_{thumb} + Loss_{smoothness} \quad (2.3)$$

adding a critical fifth component, the smoothness loss, to enforce constant velocity and prevent "jittery" movements, the system calculates the derivative of the joint angle vector across all fingers at each time step t. But still sometimes the output can appear less fluid and this is largely due to the partial observability of the initial state; without knowing where the hand was, the model must work harder to determine where it is based solely on the current muscle burst.
Also, with the transfer learning method Adaptive Batch Normalization (AdaBN), the framework can take a pre-trained model and calibrate it to the specific muscle nuances of a new subject. Remarkably, the system requires only 90 seconds of training samples from a new user to achieve a customized fit. This rapid adaptation allows the model to overcome the inter-user variability problem, eventually reaching a median accuracy on the joint angles of 6.24°.
NeuroPose also demonstrates remarkable robustness to natural variations in wrist position and mobility. The system was evaluated across five distinct configurations: normal, up, down, bend, and mobile, which includes active rotations during the task. The same dataset has been also tested with an LSTM cell that turned out to be less precise with respect to the modified U-Net architecture.

However, Putro et al. (2024) observe that, while Deep Learning models are technically capable of autonomous feature learning directly from raw sEMG data, this "end-to-end" approach requires massive computational power, especially in regression tasks. The high dimensionality of raw signals leads to architectures with an enormous number of parameters, increasing training times. This computational burden makes it extremely difficult to integrate such systems into edge devices or wearable rehabilitation tools, where power and processing resources are limited.
So they proposed a double feature crafting mechanism validated on a NinaPro subdataset (23 hand movements): first a specific feature is chosen, the median frequency is found as the best in this study, to reduce the input dimensionality and it is fed into a transformer architecture to output 22 finger joint angles, reaching a 0.970 R^2 for this regression task in a non-real-time setting, and without adding muscle fatigue issues.

A more data-driven and scalable approach to continuous hand kinematics estimation is introduced by the emg2pose benchmark, conducted by Salter et al. (2024)

[34]. The dataset's architecture is built upon massive data acquisition: over 370 hours of recordings hand motion, without taking into account the wrist, from 193 users, totaling 80 million labeled frames. Data quality is ensured by a high-level hardware setup featuring the sEMG-Reaserch Device wristband, equipped with 16 channels sampling at 2 kHz. To establish an accurate ground truth, movements were simultaneously recorded by a 26-camera motion capture rig.
The benchmark introduces vemg2pose (velocity emg2pose), a model grounded in the physiological principle that sEMG signals are inherently more coupled with dynamics (force and velocity) than with the static position of the hand.
First, a module called featurizer analyzes the raw signal, extracting salient features. These features are then processed by an LSTM decoder, which operates in an auto-regressive manner, so receiving the current features alongside the hand position estimated in the previous step. Instead of attempting to directly predict joint angles, the model predicts joint angular velocities which are integrated to update the virtual hand pose, surpassing NeuroPose in angular error also for "held-out" users, landmark distances and fluidity.

A significant advance for the regression tack is represented by the ALVI Interface (2025) [35]. Their dataset consists of 45 dynamic and 27 static daily life gestures tracked by the Oculus Quest, a markerless motion capture system, and the wireless Myo Armband. In order to maintain the same anatomy also for amputees, they used a mirroring approach for which they tracked the healthy hand and then, by reflection, the absent hand is rendered in the virtual environment.
The normalized 200 Hz raw signals are fed into a transformer based architecture called HandFormer. The efficiency of the model is enhanced by a self-supervised pre-training strategy based on a Masked Autoencoder (MAE), with 70% token masking which enables the network to learn the intrinsic structure of EMG signals prior to task-specific training. Also, instead of a costly Self-Attention, the Cross-Attention mechanism compresses the 256-point input window into 32 latent tokens. Unlike autoregressive models that suffer from error propagation, the HandFormer Decoder predicts the entire 1.28-second hand trajectory in parallel, reconstructing 20 DOFs of the fingers simultaneously. Each of the 32 learnable queries ask for one of the 32 frame of the total movement to predict.
The model achieves a remarkable correlation of 0.80 (14.50° angle error) and 0.86 (8.09° angle error) respectively in amputee and non-amputees subjects, providing high-fidelity control with a minimal latency of approximately 51 ms. A key feature of this system is its closed-loop co-adaptation: the user receives immediate visual feedback in the Virtual Reality (VR), allowing both the human and the AI, which updates its weights every 10 seconds, to mutually adapt and optimize the control strategy as highlighted in Fig. 2.4. Participant reported a growing sense of intuitive control as also amputees could quickly learn fine motor control in VR.

Few studies have approached both gesture recognition and grip strength estimation simultaneously in wearable hand-assistive robotics. Zhang et al. (2026) [36] proposed both an underactuated soft robotic glove driven by tendon-sheath artificial muscle actuators, that a transfer learning approach using a Gated Recurrent Unit based Domain-Adversarial Neural Network (GRU-DANN), which learns

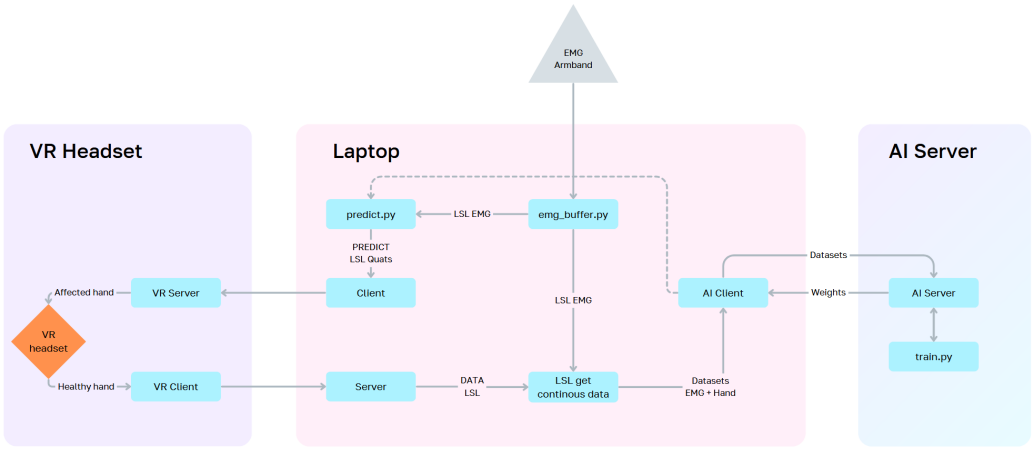


Figure 2.4 Pipeline of ALVI Interface system architecture. In their acquisition framework, data streams are synchronized using Lab Streaming Layer (LSL), a software-based system for time-aligning multimodal signals through shared timestamps.

domain-invariant features to improve cross-subject generalization without requiring user-specific retraining. DANN employs a Gradient Reversal Layer (GRL): during the backpropagation, the gradient is multiplied by a negative scalar and this inversion forces the feature extractor, the GRU, to maximize the domain classifier's loss so to deliberately "confuse" the network about the subject's identity. This adversarial process is what enables the system to maintain high accuracy when encountering a previously unseen user.
They achieved, on healthy subjects, a high within-subject accuracy of 92.12% for gesture classification and a 0.935 R^2 for grip strength estimation and for cross-subject scenarios an R^2 of 0.80. However to prioritize assistive torque there is 20.4% decrease in the total range of motion during full finger flexion. The operational framework of Zhang et al.'s system relies on a dual-stage supervised learning pipeline shown in Fig. 2.5.

In the offline training stage (purple arrows) the sEMG features, in time and frequency domain, are mapped synchronously to both gesture categories and continuous grip strength targets. While in the online control stage (black arrows) the trained model performs real-time inference, providing the myoelectric controller with immediate predictions of the user's intent. As closed-loop control strategy, the predicted grip strength is decomposed into specific assistance references for the actuators, which are then dynamically adjusted based on real-time tendon tension feedback.

Other works have explored enhancing accuracy by enriching the sensing modality itself. In this direction Spacone et al. (2025) propose a multimodal framework that fuses sEMG with A-mode Ultrasound (US) for continuous tracking of both the hand and the wrist. The hardware consists of two dry-contact armbands: an 8-channel differential EMG band placed near the elbow and a 4-channel A-mode US band positioned distally on the forearm. The framework tracks 23 Degrees of Freedom (DoF), 20 for the fingers and 3 for the wrist, using a multi-task learning approach.

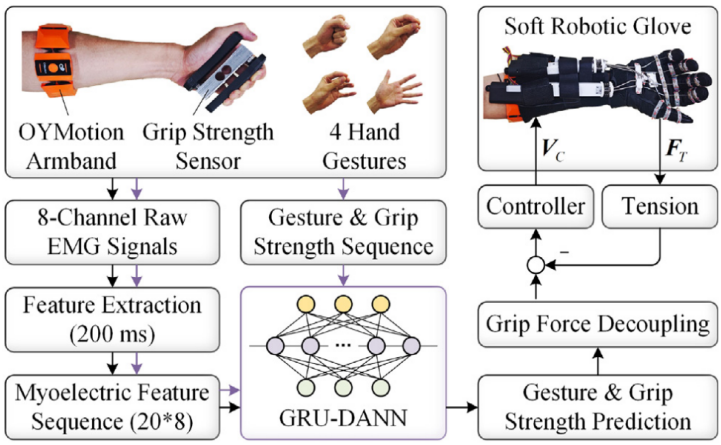


Figure 2.5 Control framework for the soft robotic glove based on gesture and grip strength simultaneous prediction.

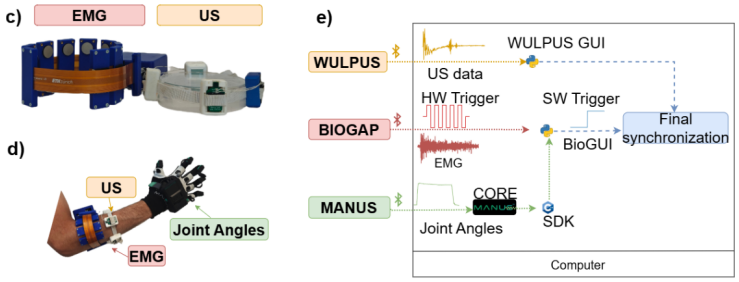


Figure 2.6 Ultrasound signals are acquired using WULPUS, the dedicated US acquisition interface, while data from the MANUS glove are streamed to MANUS Core and then collected by BioGUI, a software that centralizes and synchronizes EMG and glove data. EMG signals and the hardware synchronization trigger are provided by BioGAP, the EMG acquisition hardware. BioGUI first synchronizes BioGAP and MANUS data via a software trigger, then aligns them with the WULPUS signals using the hardware trigger.

The model utilizes a lightweight encoder-decoder architecture with a late-feature fusion strategy: instead of combining EMG and US signals at the input level, each modality is first processed independently through separate encoder branches. The resulting high-level features are then flattened, concatenated, and processed by a Multilayer Perceptron (MLP) head specific to each task, 20 DoF of the hand and 3 DoF of the wrist. Here the third DoF for the wrist represents the ulnar deviation and supination.
One of the primary technical challenges in multimodal systems is the alignment of heterogeneous data streams. As illustrated in Figure 2.6, the authors developed a rigorous synchronization pipeline based on both software and hardware triggers to ensure temporal consistency across EMG, ultrasound, and glove data.

The framework's core innovation lies in its Late-feature Fusion architecture, designed to handle the simultaneous prediction of hand and wrist kinematics. As shown in Figure 2.7, the model processes EMG and US data through separate,

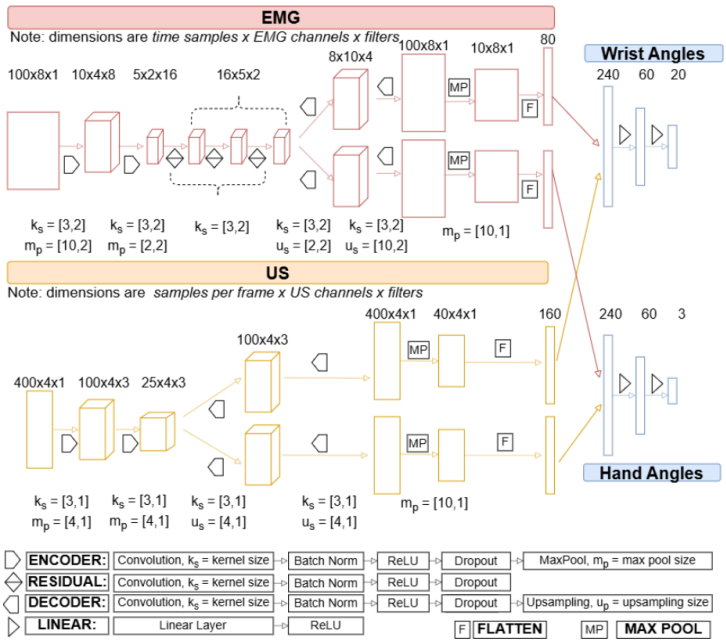


Figure 2.7 Network Architectures for EMG and US, with late-feature fusion

lightweight encoder branches. These features are then flattened and concatenated into a shared representation.
A key strength of this design is the use of Multi-task Learning: rather than using a single output head, the network employs distinct Multilayer Perceptron (MLP) heads dedicated separately to the 20 DoFs of the fingers and the 3 DoFs of the wrist. This separation allows the model to better specialize in the different biomechanical behaviors of the hand and the forearm joints. 2.7.
The evaluation of the system's performance in real-world scenarios shows that this fusion has higher resutls than the single models, but they still only achieved R^2 of 0.61 and an RMSE of $10.6° \pm 2.0°$.

2.2.5 Motor Redundancy and the Synergy Hypothesis

The extraordinary mechanical complexity of the human hand presents a significant computational challenge for the CNS, known as the musculo-skeletal redundancy problem (Bernstein, 1967) (Grinyagin et al., 2005, [37]). To simplify control, the CNS is hypothesized to operate through synergies, functional units that coordinate multiple elements simultaneously. This is further supported by cortical evidence. As demonstrated by Leo et al. (2016) [38], hand postures can be reliably reconstructed from a functional magnetic resonance imaging (fMRI) brain activity by mapping neural signals onto kinematic synergies.
Research distinguishes between:

1. Static Postural synergies: refer to the hand angular configuration during a final position, like a gesture or a grasping pose (Santello et al.,1998, [39]).

2. Kinematic synergies: describe the coordinated covariation of joint angles

during reach-to-grasp tasks as first studied by Santello et al. (1998) [39] on 57 subjects who demonstrated that three hand postural synergies were identified through a principal component analysis (PCA) that accounted for the 84% of variance in the kinematic data and characterized hand configurations as linear combinations of finger joints:

(a) First synergy (PC1): modulates the abduction/adduction and flexion/extension of all finger joints (both proximal and distal). This component accounts for the largest variance in hand shape, leading to a posture characteristic of a power grasp.

(b) Second Synergy (PC2): reflects thumb opposition combined with the flexion-extension of the distal joints only. This synergy is primarily associated with precision movements (pinch) and the manipulation of small objects.

(c) Third Synergy (PC3): specifically involves the flexion and opposition of the thumb, optimized for grasping flatter or wider objects.

More recent large-scale studies have expanded this framework. Karque-Bou et al. (2019) [40], analyzing 77 subjects performing 20 different grasps from ADL, found out through PCA and hierarchical clustering that twelve synergies are required to account for more than 80% of the overall kinematic variation. The new nine synergies are crucial for fine motor control, such as the independent control of the thumb and index finger, which are essential for skilled manipulation.

3. Kinetic (or dynamic) synergies: describe the coordination and covariation of contact forces and torques during hand stabilization and object manipulation. Grinyagin et al. (2005) [37] studied the kinetic synergies during human precision-grip movements and found out that they are remarkably simple and consistent across all subjects compared with the synergies found for movement kinematics. In fact, at the level of movement kinematics, only for large aperture of the hand the angular contributions are subject-invariant.
Also, in all subjects and conditions, the torques in the proximal joints contributes most significantly to the total variance of the movement dynamics, whereas distal torques only rotate the distal phalanges. In general the CNS, to ensure functional success in such tasks, employs three primary strategies to adapt joint torques:

   (a) Linear Scaling: as the required grip aperture increases, the CNS achieves adaptation through a simple linear increase in joint torques, maintaining the same relative contribution from each joint.

   (b) Torque Synchronization: as movement velocity increases, the synchronization between the time courses of joint torques becomes more pronounced, leading to higher correlations.

   (c) Motor Equivalence: high adaptability by redistributing effort between digits.

4. Muscle synergies (MS): while kinematic and kinetic synergies describe the mechanical output, muscle synergies represent the CNS control strategy within the sEMG space. Tagliabue et al. (2015) [41] showed that the principal muscle synergy, from PCA, exhibit a phase lead of up to 50 ms relative to the principal kinematics synergy during two-digit reach-grasp-hold tasks, suggesting that muscle synergies are at least partially at the source of kinematic synergies.
   This study also showed that both kinematic and muscle synergies can simultaneously account for grip configuration and grip force. Moreover, the reach phase already contains anticipatory information about the upcoming grasp, indicating a feedforward control strategy.
   Berger and d'Avella (2014) [42] showed that a small set of 4–5 muscle synergies is sufficient to control hand force in multiple directions, achieving the same accuracy as with individual muscle control.

   To mathematically extract MS from raw sEMG signals, Non-negative Matrix Factorization (NMF) is used over PCA, because muscle activations are inherently non-negative, as muscles can only pull (contract) and never push.
   NMF requires that after the typical pre-processing introduced in 2.1.1, the signal is low-pass filtered and rectified in order to obtain the sEMG envelopes, which are then normalized to reduce inter-muscle amplitude differences and arranged into a matrix containing the activity of all recorded muscles over time. The resulting matrix constitutes the input V for the NMF algorithm and it is formally modeled as:

$$\boldsymbol{V \approx W \cdot H}, \quad \boldsymbol{W, H \geq 0} \tag{2.4}$$

   where $V \in R^{m \times t}$ is the original EMG matrix (muscles $\times$ time), $W \in R^{m \times n}$ contains the muscle weights (spatial synergies), and $H \in R^{n \times t}$ describes the temporal activation patterns of the n identified synergies. NMF is an iterative process that minimizes the Euclidean distance between the original signals V and the reconstruction obtained by the multiplication of W and H. Typically the reconstruction of the original dataset is quantified with the coefficient of determination R^2.
   The processing pipeline to extract MN is illustrated in 2.8.

5. Motoneuron synergies (MNS): as Tanzarella et al. (2021) [43] report in their work, because muscle synergies are extracted from the sEMG signal, they only indirectly reflects the neural drive received by a muscle, therefore it carries information on the overall activity of the innervating motor neurons, effectively reducing each muscle to a single dimension of control.
   Also, the number of spinal motor neurons is orders of magnitude greater than the number of muscles so the EMG envelopes may fail to capture the true complexity of neural inputs and the real correlations between motor neurons. Subsets of motor neurons within the same motor pool might receive distinct neural commands that are "masked" when averaging the signal into a linear envelope.

This highlights the importance of moving beyond global EMG amplitudes toward the decomposition of individual motor units to accurately identify the high-dimensional control strategies of the CNS. This decomposition is achievable through HD-sEMG analysis.

Tanzarella et al., by recording with two 64-channel grids the activity of dozens of motor neurons innervating 14 intrinsic and extrinsic hand muscles during various isometric grasps, i.e. without any significant change in muscle length, their research identified four motoneuron synergies, accounting for 70% of the variance of the activity of 54.1 ± 12.9 motor neurons, and four functionally similar muscle synergies. But they found that the motor neuron synergies are more effective in discriminating the forces applied by each finger.

Also, Capsi-Morales et al., provide experimental evidence that MNS map directly onto the kinematics synergies of the hand, significantly outperforming traditional EMG synergies in dynamic tasks, so with a continuous modulation of muscle length to reach, shape, and manipulate objects.
Motoneuron synergies capture the "pure" neural command, remaining consistent across different sessions and subjects. This allows them to cover over 90% of the hand's mechanical manifold, enabling the control of infinite postures, including in-hand manipulation, through a reduced number of common activation primitives.
The target hit rate for specific hand postures was higher compared with muscle synergies, with the difference being particularly pronounced for prosthesis users with a success rate of 82.5% compared to only 35% reached with traditional envelope-based synergies.
The number of synergies was estimated from the R^2 curve: MNS (~ 6) have a greater dimension than MS (~ 2 in different sensors' configurations so to avoid that their spatial distribution affects the results) in explaining the same amount of variance (80%).
When matched in number the MN synergies showed a higher degree of orthogonality than muscle synergies. Last, not all the MN synergies could be matched with corresponding muscle synergies.
This superiority is further supported by the principle of motor anticipation. As noted by Chen et al. (2020) [15], the neural drive physically precedes the mechanical action, making pose prediction more robust and responsive compared to global EMG features.

In order to extract motoneuron synergies, the input matrix for the NMF algorithm follow the same algebraic structure as the one used for muscle synergies (Eq. 2.4), but the meaning of the three terms shifts from a macroscopic muscle-level to a microscopic neural-level:

$$\boldsymbol{S \approx W_{MN} \cdot H_{MN}} \tag{2.5}$$

where matrix $S \in R^{k \times t}$ is the Smoothed Discharge Rates (SDRs) of the k identified motor units. $W_{MN} \in R^{k \times n}$ contains the motoneuron weights for

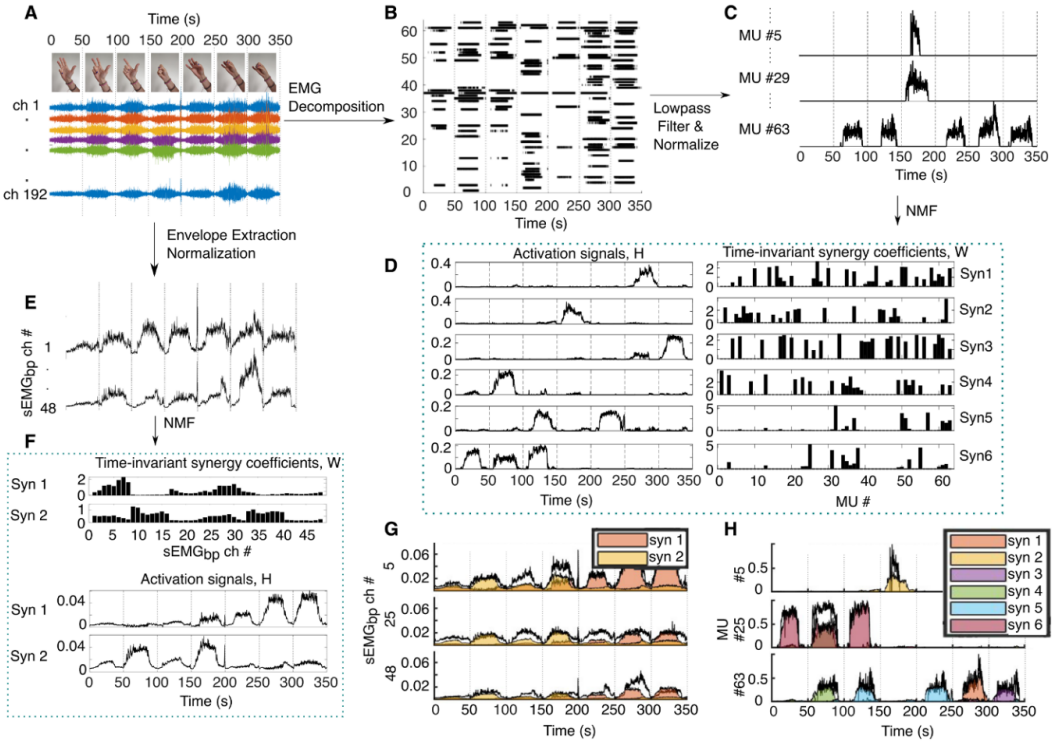


Figure 2.8 Muscle/Motoneuron Synergy extraction pipeline. In the study of Capsi-Morales et al. (2025) [10] high-density EMG signals were recorded using 192 monopolar electrodes (A). These signals were first converted into 48 bipolar EMG channels in order to improve signal quality and reduce common-mode noise. After band-pass filtering, rectification and envelope extraction (E), the EMG envelopes were normalized and used as input for the NMF decomposition. The NMF decomposition produces a set of time-invariant muscle synergy coefficients W and the temporal activation signals H (F). The reconstructed signals matches the original rectified EMG signals demonstrating that a small number of muscle synergies can accurately represent the observed muscle activity (G). The high-density signals are first decomposed into the activity of individual MU. This results in raster plots that represent the precise discharge times (spikes) of the decoded motor neurons (B). spike trains are then smoothed using a low-pass filter and normalized to obtain SDRs (C). The SDRs of the identified motor units are arranged into the input matrix S and processed via the NMF algorithm to find time-invariant MN synergy coefficients W and temporal activation signals H (D). the reconstructed motor unit activation patterns are compared against the original SDRs visualized as solid black lines (H).

the n neural synergies, and $H_{MN} \in R^{n \times t}$ represents the corresponding neural temporal activation patterns.

In the present work, the activation matrix H is used as the input representation for the mapping between neuromuscular activity and hand kinematics. By projecting high-dimensional EMG-derived motor unit activity onto a low-dimensional synergy space, the input is reduced while preserving task-relevant neural information.

2.3 Decoding Motor unit activity from HD-sEMG signals

While the Convolutive EMG mixing model provides a formal description of how sEMG signals are generated, the practical extraction of motor unit spike trains from HD recordings requires solving a challenging inverse problem.

Blind Source Separation (BSS) techniques, including Convolution Kernel Compensation (CKC) and Independent Component Analysis (ICA), became dominant framework for EMG decomposition.

The Convolutive EMG mixing model can be written in its matricial form:

$$\boldsymbol{x(n) = H\hat{s}(n)} \tag{2.6}$$

where $\mathbf{H}$ is the kerne matrix containing time-shifted versions of the MUAPs and $\boldsymbol{s(n)}$ is the accordingly extended source representation of the MUSTs, to facilitate the application of BSS techiniques, as shown in Fig. 2.9.
Within this framework, CKC aims at estimating the motor unit spike trains without explicit knowledge of the mixing matrix $\mathbf{H}$.
The method assumes that $\mathbf{H}$ is stationary over time and exploits the sparsity and temporal independence of motor unit spike trains, as well as the condition that the number of observations exceeds the number of active motor units.
The main steps of the CKC can be summarized as follows:

- Estimate the correlation matrix of the observations $\mathbf{C}_x = E[\mathbf{x}(n)\mathbf{x}^T(n)] = \mathbf{H}\mathbf{C}_{\hat{s}}\mathbf{H}^T$ from the EMG signals. It quantifies the similarity between EMG channels over time. Each element of that matrix measures the correlation between a pair of channels.

- Estimate the cross-correlation vector $\boldsymbol{c_{s_i,x}}$ between the spike train of the i-th motor unit and the EMG signals, capturing how strongly the activity of that motor unit is reflected in each channel. The cross-correlation vector $\boldsymbol{c_{s_i,x}}$ cannot be directly computed since the true spike trains are unknown. Therefore, it is estimated using an initial approximation of the motor unit activity, for instance by identifying candidate discharge times and computing the average EMG response aligned to these events.

- Compute the separation filter as $\boldsymbol{c^T_{s_i,x}\mathbf{C}_x^{-1}}$.

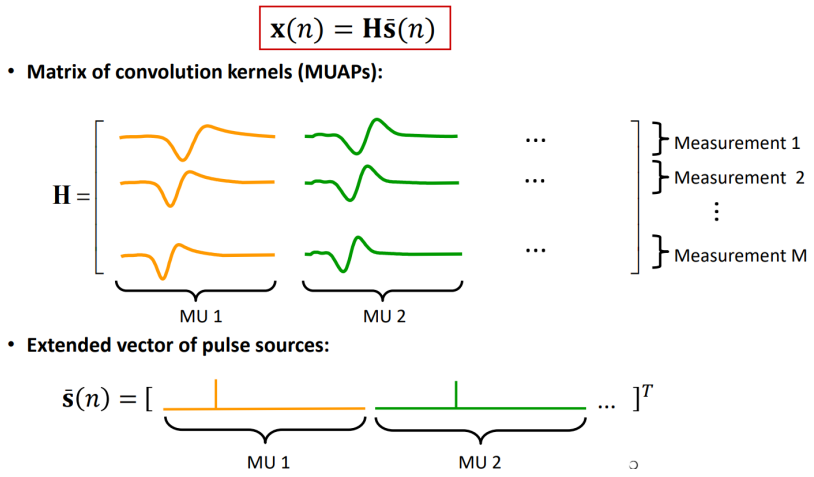


Figure 2.9 Matricial representation of Convolutive EMG Mixed Model [44]

- Apply the filter to the observations to obtain the estimated spike train $\hat{s}_i(n)$:

$$\hat{s}_i(n) = \boldsymbol{c}_{\boldsymbol{s_i,x}}^{\boldsymbol{T}} \mathbf{C}_{\boldsymbol{x}}^{-\mathbf{1}} \boldsymbol{x(n)} \tag{2.7}$$

  The procedure is typically stopped based on signal quality criteria, such as the absence of additional identifiable spike trains or a decrease in the signal-to-noise ratio (SNR) of the extracted components.

However, the estimation of the separation vector in CKC relies primarily on second-order statistics, such as correlation and covariance, which capture linear dependencies between signals.

In contrast, ICA-based methods explicitly exploit higher-order statistics to enforce statistical independence between sources. These include measures of non-Gaussianity which allow distinguishing signals even when they are uncorrelated but not independent.
As a result, ICA methods iteratively update the separation vectors to maximize independence, leading to more flexible and robust source separation strategies.

In practice, these decomposition frameworks are often implemented through dedicated software environments that support both automatic and semi-supervised processing.
Tools such as the open-source MUedit (Avrillon et al. [45]) enable the application of BSS-based algorithms, FastICA in this case, to high-density EMG signals, providing interactive visualization of motor unit pulse trains and allowing manual refinement of discharge times.
To better illustrate the underlying mechanism of FastICA, the separation process can be interpreted as the iterative estimation of a separation vector for each motor unit as shown in Fig. 2.10.

Similarly to CKC, FastICA relies on an extended representation of the EMG signals which are then whitened, such that their covariance matrix becomes the identity, simplifying the separation problem.

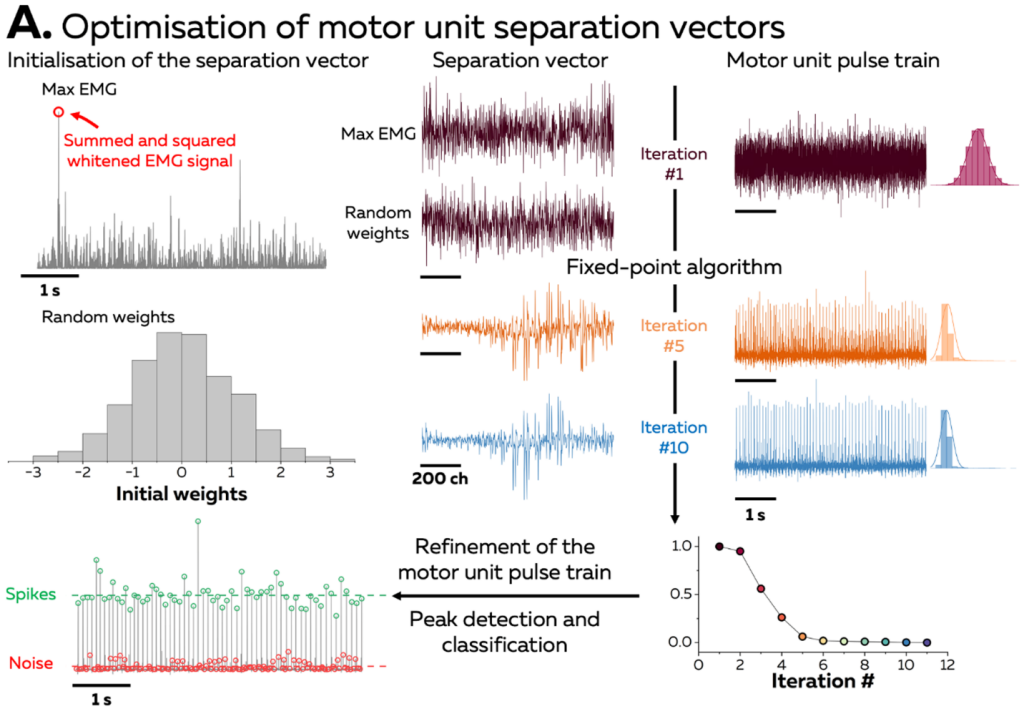


Figure 2.10 FastICA Algorithm of the MUedit framework.

Starting from an initial guess, the algorithm applies a fixed-point optimization scheme that updates the separation vector w so as to maximize the non-Gaussianity of the projected signal defined as

$$y = w^T x \tag{2.8}$$

where x is the whitened EMG signal. This projection corresponds to the motor unit pulse train, from which discharge times can be identified.
Unlike standard gradient descent, which requires a learning rate and can be slow to converge, the fixed-point algorithm finds the direction of maximum non-Gaussianity by searching for the zeros of the derivative g of a contrast function (e.g., *tanh* or *skew* or *kurtosis*) .
The update rule follows the form:

$$w_{k+1} = E\{xg(w^T x)\} - E\{g'(w^T x)\}w \tag{2.9}$$

To ensure numerical stability and prevent the divergence of the weights, the separation vector is normalized after each iteration.
This specific mathematical structure ensures cubic or quadratic convergence, making it exceptionally fast.

The separation vector can be initialized either randomly or using the sEMG signal values at time instants of maximal global activity (Max EMG), which can accelerate convergence. The fixed-point iterations continue until the variation of

the separation vector between successive iterations falls below a predefined threshold.

At convergence, the estimated pulse train exhibits a sparse distribution, where high amplitude peaks correspond to motor unit discharges. These spikes are then identified through peak detection and clustering procedures, allowing discrimination between true discharges and background noise.

To assess MU reliability, two primary metrics are used for this validation: the Silhouette Value (SIL), that measures the normalized distance between the spikes and the noise and can be used to assess the reliability of the motor unit pulse trains (Negro et al.) and the Coefficient of Variation of Inter-Spike Intervals (CoV_{ISI}), the standard deviation of Inter-Spike Intervals normalized by their mean (Kaifosh et al.).

The authors also propose extensions towards online processing, such as the I-SPIN framework (Rossato et al., 2024, [46]), which enables real-time motor unit tracking by integrating adaptive decomposition strategies.

To overcome the limitations of stationary approaches and compensate for problems in dynamics, such as those introduced by changes in muscle geometry during movement, adaptive decomposition methods have recently been proposed such as the one of Mendez Guerra et al. (2024) [47]. Their method continuously adapts the separation parameters over time. By doing so, the algorithm is able to track changes in the EMG mixing process enabling real-time tracking of motor unit activity with high accuracy across dynamic conditions, representing a significant step toward robust neural decoding.

Despite the increasing availability of adaptive and online decomposition methods, offline frameworks such as MUedit remain widely used due to their robustness, ease of use, and availability of validated processing pipelines.

In this work, although the dataset involves dynamic conditions, a FastICA-based decomposition implemented in MUedit is adopted to ensure a controlled and reproducible extraction of motor unit activity. This choice is motivated by the need to define a clear and accessible processing pipeline, which will serve as the basis for subsequent analysis of motor unit synergies and their use in predictive models of kinematic behavior.

2.4 HD-sEMG to hand pose mapping

Recent studies have explored how these neural signals can be integrated to predict continuous hand movements. By combining decoded motor unit activity with biomechanical or synergy-based models, researchers aim to translate high-resolution neural information into functional hand pose estimation.

Chen et al. (2020) used a high-resolution array of 192 channels distributed across

the forearm muscles to record 11 different hand and wrist gestures.
By applying the CKC algorithm, they were able to find 29 MU per movement with a decomposition accuracy of over 90%. A SVM categorizes each individual motor unit into one of the eleven movement groups. Then through a Voting Machine algorithm, every time a motor unit fires an impulse, it casts a "vote" for the gesture it was assigned to. But this system remains anchored to discrete classification and it is performed only offline.

Building on these initial findings, Tacca et al. (2024) further explored how HD-sEMG can be leveraged for continuous hand pose regression, moving beyond discrete gesture classification to reconstruct complex joint kinematics.
They introduced NeuroLife EMG sleeve, a high-density (HD-EMG) interface equipped with up to 150 electrodes (configured as 75 bipolar channels). The study demonstrated the possibility of mapping this dense electrical information to a highly complex kinematic model.
To obtain the necessary kinematics for training the regression models, participants wore the CyberGlove III, an 18-sensor motion capture glove capable of calculating 23 distinct joint angles, and performed 37 different hand movements and grasps for sequential tasks and 19 of them in random sequences.
As illustrated in Fig. 2.11, the processing pipeline transforms filtered EMG signals into lagged Root Mean Square (RMS) features, which are then flattened and fed into a linear NN to predict joint angles.
The system achieved a remarkable coefficient of determination ($R^2 = 0.884$) for continuous joint angle prediction in sequential tasks, but when the user performs random movements the performance decreases with $R^2 = 0.750$.
Despite this, the MAE was kept below 10° for all joints, with a grand average error as low as $1.8° \pm 0.04°$ and $3.4 \pm 0.07°$ respectively for sequential and mixed tasks. Also for a classification task the accuracy varies between 88.5% and 99.3%.

Capsi-Morales et al., as previously introduced, took a step further by integrating decoded neural drive with mechanical synergies to enable fluid, real-time control of a prosthetic hand, called SoftHand Pro-2. It features 2 DoA, 15 DoF achieved through a tendon-driven transmission system that distributes force across all fingers which allows power and precision grasps.
During the offline analysis nine participants performed seven grip types or hand postures with isometric forces, with the elbow flexed 90° to rest the weight of the arm on a table with wrist and hand area free to move. The grips included the flexions of the fifth digit or little finger (p_1), the fifth and the ring fingers(p_2), or the fifth, ring, and middle fingers(p_3); flexion of all digits to apply force and make contact with the palm of the hand (p_4, power grasp); or making a contact with the thumb in opposition with the index finger ((p_5, pinch), the index and middle fingers (p_6, tipod), and index to ring fingers (p_7) The MVC for each grip is computed and participants followed a target on a screen with a positive ramp (1% MVC/s), followed by a 20-s isometric contraction at 15% of MVC and a final negative ramp (1% MVC/s).
Following the experimental protocol, the extraction of motor neuron synergies relies on a signal processing pipeline conceptually aligned with the framework proposed

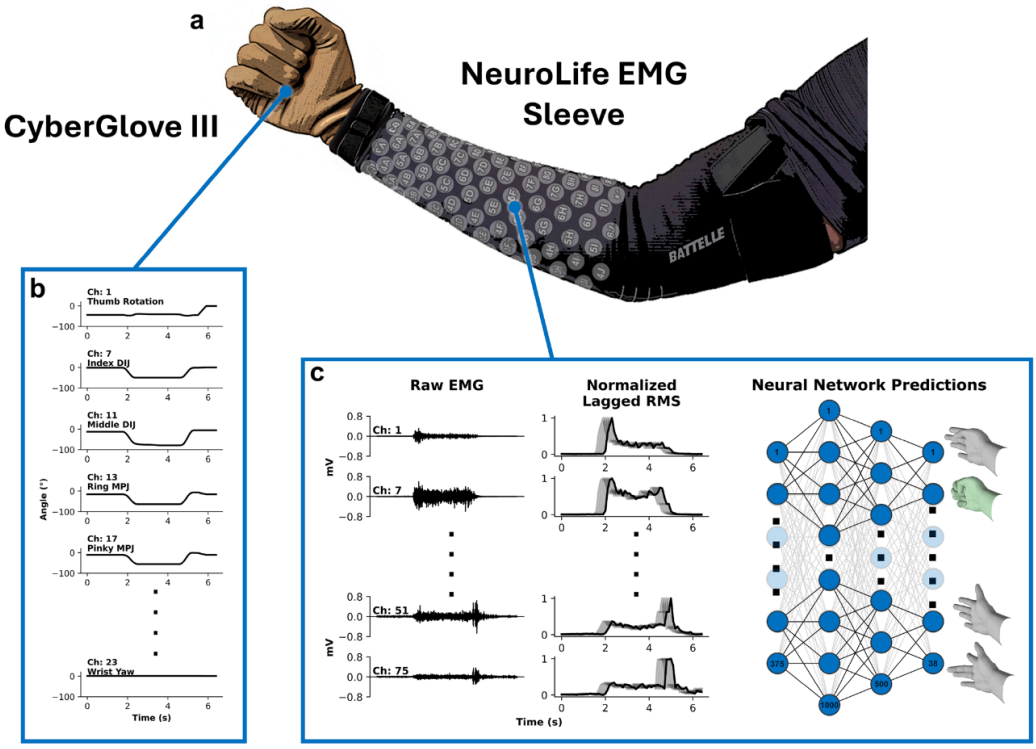


Figure 2.11 High-resolution decoding with the NeuroLife EMG sleeve

by Tanzarella et al. (2021), as illustrated in Fig. 2.12.
high-density sEMG signals were recorded using 192 monopolar channels over a total duration of approximately 350 seconds, across all grip types. To ensure consistent identification and tracking of motor units across different tasks, the EMG signals corresponding to the various grip conditions were concatenated prior to decomposition. To ensure consistent identification and tracking of motor units across different tasks, the EMG signals corresponding to the various grip conditions were concatenated prior to decomposition.
Unlike the approach in Tanzarella et al., where CKC was employed, Capsi-Morales et al. adopted the BSS method based on the fixed-point algorithm, following the framework introduced by Barsakcioglu and Farina (2018) [48].
Once the motor unit discharge timings were identified, the spike trains were converted into SDRs. These neural signals were then used as input to 100 iterations of the NMF repeated 10 times for each number of synergies, from one to 9, with different seeds per repetition for the matrices $\boldsymbol{H}$ and $\boldsymbol{W}$, with reconstruction values already reported in 2.2.5.

Neural activation signals (H), extracted by the NMF, are directly mapped into a continuous space of hand postures (P) as schematized in Fig.2.13. To achieve this, a set of trained hand postures is defined as:

$$\boldsymbol{P} = \boldsymbol{K}\boldsymbol{H} \tag{2.10}$$

where $K \in R^{T \times S}$ is a weight matrix, with T representing the number of hand

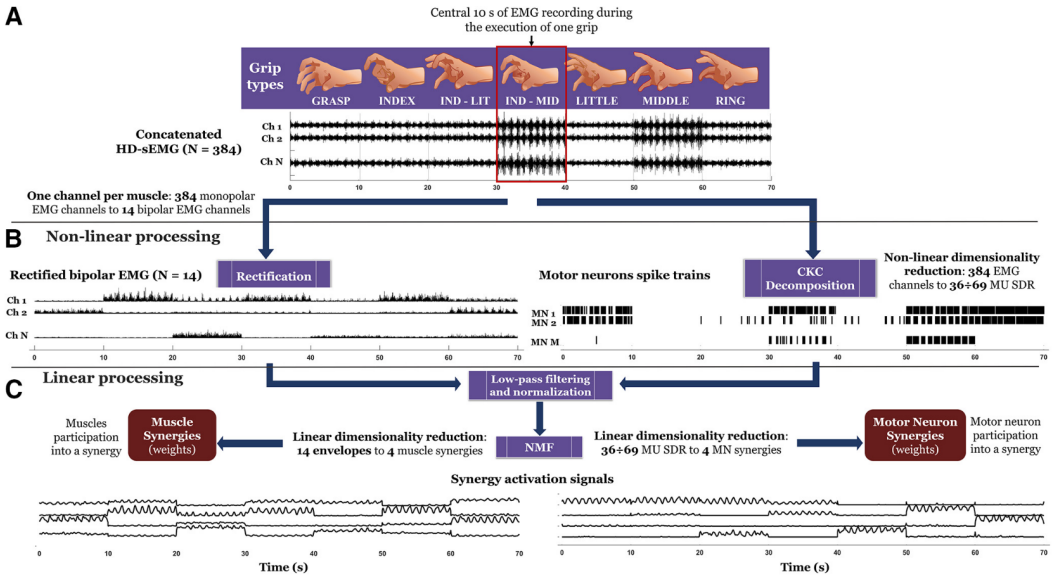


Figure 2.12 sEMG decomposition and NMF pipeline. For each grip type (top), the central 10 s of EMG recordings was concatenated to obtain 70-s long EMG signals (10 s per grip type). In the first branch, a reduced set of bipolar EMG channels was obtained and rectified to generate conventional EMG envelopes. In the second branch, the HD monopolar signals were decomposed using a BSS approach to extract motor unit spike trains. The resulting signals were then low-pass filtered to obtain smooth temporal profiles, representing either global muscle activation or neural drive. Finally, NMF was applied to these signals to extract muscle synergies and motor neuron synergies, respectively, providing a low-dimensional representation of the underlying control structure [43].

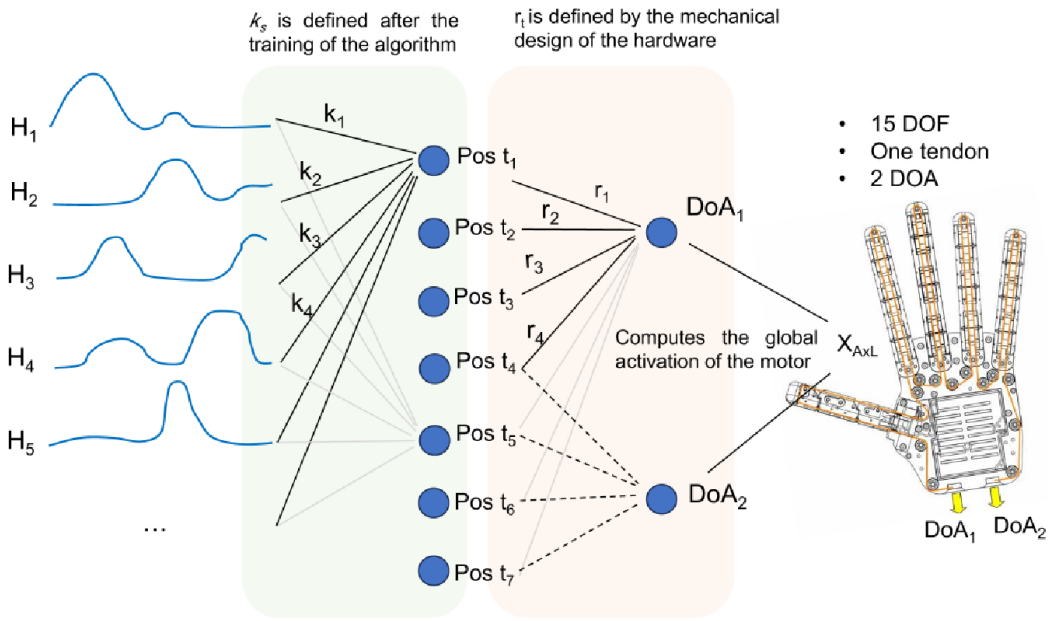


Figure 2.13 Mapping of MN Synergies and Postural Synergies for SoftHand Pro-2.

	CapgMyo	SEEDS	GrabMyo	Hyser	CEMHSEY
No. Subjects	10–18	25	43	20	13 (GRASP), 6 (GESTURE)
Amputees?	No	No	No	No	No
No. Movements/Gestures	8, 12, 2	13	16	34	7 (GRASP), 11 (GESTURE)
No. Electrodes	128 (HD)	128 (HD) +8	28	256 (HD)	320 (HD)
Kinematics or Force?	No	Kinematics	No	Finger Force	Hand-Gripping & Finger Force (GRASP), No (GESTURE)
Multi-Day?	2 days	No	3 days	2 days	11 days
Consecutive Days?	No	\	No	Yes	Yes

Figure 2.14 Comparison between some open-access sEMG datasets (Yang et al., 2025, [49]).

tasks.
During the training phase, the weights k_s within the matrix K are initialized based on the mean activation of each corresponding synergy during a specific task. To ensure reliability, a synergy is only considered "active" if its signal exceeds a threshold of 10% of maximum activity for more than 40% of the task's duration. These weights are then normalized for each posture so that their sum equals 1.
The trained postures are then converted into the two DoAs available in the multi-synergistic prosthetic hand used, $X \in R^{A \times L}$ (where A is the number of DoAs):

$$\boldsymbol{X = RP} \tag{2.11}$$

2.4.1 sEMG to hand pose dataset

As shown in Fig. 2.14 there are few open-access sEMG datasets comprising HDsEMG signals. Specifically, CEMHSEY (Yang et al., 2025, [49]) includes finger force and

hand-gripping data (the global force exerted by the whole hand during a grasp), through hand grip or pinch dynamometer, recorded in 11 consecutive days with 320 channels of HDsEMG and validated with the usability of force regression, hand gesture recognition, and motor unit decoding.

In contrast, SEEDS (Matran-Fernandez et. al, 2019, [50]) stands out as one of the few datasets providing a mapping with HDsEMG and hand kinematics, wrist and fingers (with the 18-sensor data-glove CyberGlove III), rather than force, with the possibility to apply regression methods and extraction of the neural drive, but is limited to within-day recordings.

2.5 Feedback in Rehabilitation and Myoelectric Control

Beyond signal decoding, the clinical effectiveness of the rehabilitation process is strictly tied to patient engagement.

As shown by Amin et al. (2024) [51], fully immersive VR-based games, in Unity3D with Oculus Quest 2, that incorporate cognitive engagement and visual feedback, lead to significantly better and more enduring motor recovery compared to conventional therapy, especially for complex tasks like pinching or grasping. In particular, for subacute stroke patient type, VR acts on a phase in which brain's neuroplasticity is adaptable to rewiring and modifications, leading to marked improvements in manual dexterity and strength, as evidenced by significant increases in different clinical tests.

Beyond the visual and cognitive engagement provided by VR, closing the control loop in myoelectric prostheses requires addressing the limitations of sensory feedback in realistic environments.

As demonstrated by Tchimino et al. (2023) [52], providing artificial somatosensory feedback is essential when dealing with compliant (deformable) objects, where precise force regulation is critical. Their study shifts the focus from external sensing to internal state awareness by conveying the intensity of the user's own muscle contraction through real-time tactile stimulation. Unlike traditional systems that provide feedback based on grasping force measured at the fingertips, information that can often be redundantly estimated through incidental visual cues like object deformation, this EMG feedback provides an "intrinsic" measurement of the controller that remains otherwise imperceptible to the user.
Physically, as shown in Fig. 2.15, this is achieved through an array of four vibrotactors positioned on the upper arm, where the sEMG amplitude is discretized into five levels, each corresponding to a specific vibration pattern. This discretization allows the user to "feel" the magnitude of the command being sent to the prosthesis. Crucially, because the sEMG signal in proportional control directly dictates closing speed and final force, this feedback enables a predictive control strategy. The user can modulate their contraction and anticipate the final grip intensity even before the prosthesis makes contact with the object. This system provides constant tactile

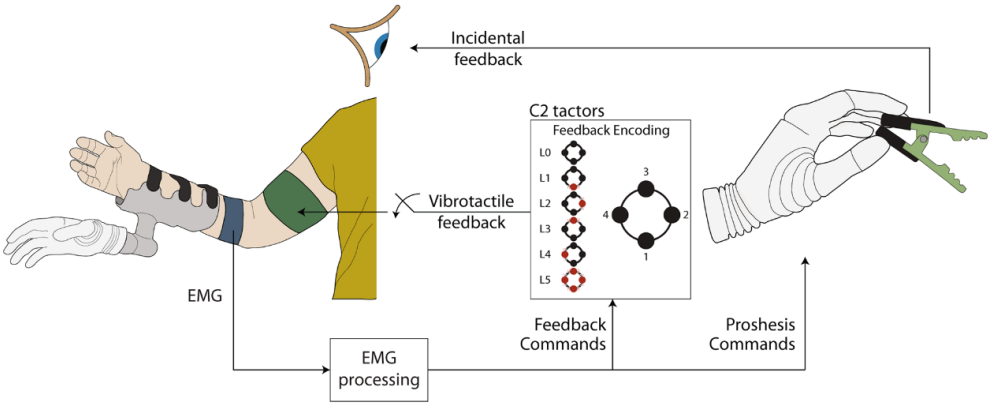


Figure 2.15 The user receives discretized vibrotactile stimulation corresponding to the muscle activation level, enabling precise force regulation through a predictive control strategy.

confirmation that stabilizes the human-machine interface and ensures refined force regulation regardless of the availability of visual or auditory cues.
Although this study focuses on prosthetic control, the concept of intrinsic EMG feedback can be effectively adapted to stroke rehabilitation with lost of proprioceptive information.

Chapter 3

Methodology

This chapter details the pipeline developed to extract the neural drive information from the sEMG as the primary control signal, following the definition of an architecture which maps decoded neural dynamics to hand kinematics, enabling a fluid and intuitive interaction within the virtual environment.
The methodology is structured into three main phases.
First, investigate the motoneuron decomposition process to extract MUSTs and their corresponding SDRs. This stage leverages State Of The Art open source tools to decode the firing patterns of individual motor units from HD-sEMG recordings.
Second, address the high dimensionality of the neural signal by identifying neural synergies. By applying dimensionality reduction techniques, extract the latent components, which represent the temporal evolution of the neural drive. These synergies serve as the "neural manifold" that provides a more robust and biologically plausible input for continuous decoding.
Finally, these neural synergies are fed as the continuous input to an LSTM model which predicts hand poses.

3.1 Preliminary Experimental Setup

Before selecting the dataset adopted in this work, an initial objective of the internship carried out at "Fondazione Santa Lucia IRCCS" was to develop an experimental protocol for the simultaneous acquisition of HD-sEMG and hand kinematic data and subsequent motor unit decomposition and neural synergy analysis.

The integration between the MANUS Quantum Mocap Metagloves system [53] and the Movella Xsens motion capture system [54] was investigated. The available MANUS Core + Xsens license allows either the acquisition of hand kinematics through the MANUS software with subsequent data export in CSV format or the simultaneous recording of whole body kinematics from the 18 Xsens inertial sensors together with hand kinematics acquired by the MANUS gloves through the Xsens software.

A further aspect of the preliminary investigation concerned the synchronization between the motion capture system and the HD-sEMG acquisition hardware.

During the internship, the EMG-USB2+ amplifier (OT Bioelettronica, [55]) equipped with 64-channel high-density electrode grids was available, enabling the acquisition of HD-sEMG signals suitable for motor unit decomposition. Signal acquisition and visualization were performed using the OTBioLab+ software environment [56], which provides tools for electrode configuration, real-time signal monitoring, data recording, and offline processing.

Unlike many experimental setups reported in the literature, which rely on the Lab Streaming Layer (LSL) [1], a protocol to synchronize heterogeneous devices, the EMG amplifier used in this work does not support LSL communication.

Consequently, synchronization was planned through a hardware trigger. Specifically, the auxiliary input of the amplifier was configured as an AUX Trigger by selecting Direct Connection to Auxiliary Input within the acquisition software (Fig. 3.1), enabling the reception of a TTL synchronization signal from the Xsens system (Fig. 3.2) through a BNC connection. This configuration would have allowed the temporal alignment between the electrophysiological and kinematic recordings.

The OTBioLab+ software also includes an integrated offline decomposition module (Decomponi) capable of identifying motor units from HD-sEMG recordings and providing their corresponding spike trains, instantaneous discharge rates (IDR), innervation pulse trains (IPT), and manual editing tools for validating the detected motor units. The software allows the operator to review the identified motor units according to quality metrics, approve or reject individual units, and export the validated spike trains for further analysis.

Although this decomposition workflow was evaluated during the preliminary phase, it was not adopted in the present work. Instead, the objective is to develop a processing pipeline based entirely on open-source algorithms for HD-sEMG decomposition. This choice was motivated by the greater flexibility of open implementations, which allow complete access to the decomposition pipeline and represent a more suitable starting point for future developments toward real-time motor unit decoding and neural interface applications.

3.2 Dataset

The dataset used in this study is the HD-FW KIN dataset introduced by Guo et al. [58]. It provides synchronized HD-sEMG recordings and raw hand sensor measurements, allowing the investigation of the relationship between neuromuscular activity and hand motion.

Before considering the current dataset, the SEEDS dataset (Matran-Fernandez et al. [50]) was also explored. However, the recordings for each movement in the

[1] An open-source networked middleware ecosystem to stream, receive, synchronize, and record neural, physiological, and behavioral data streams acquired from diverse sensor hardware [57].

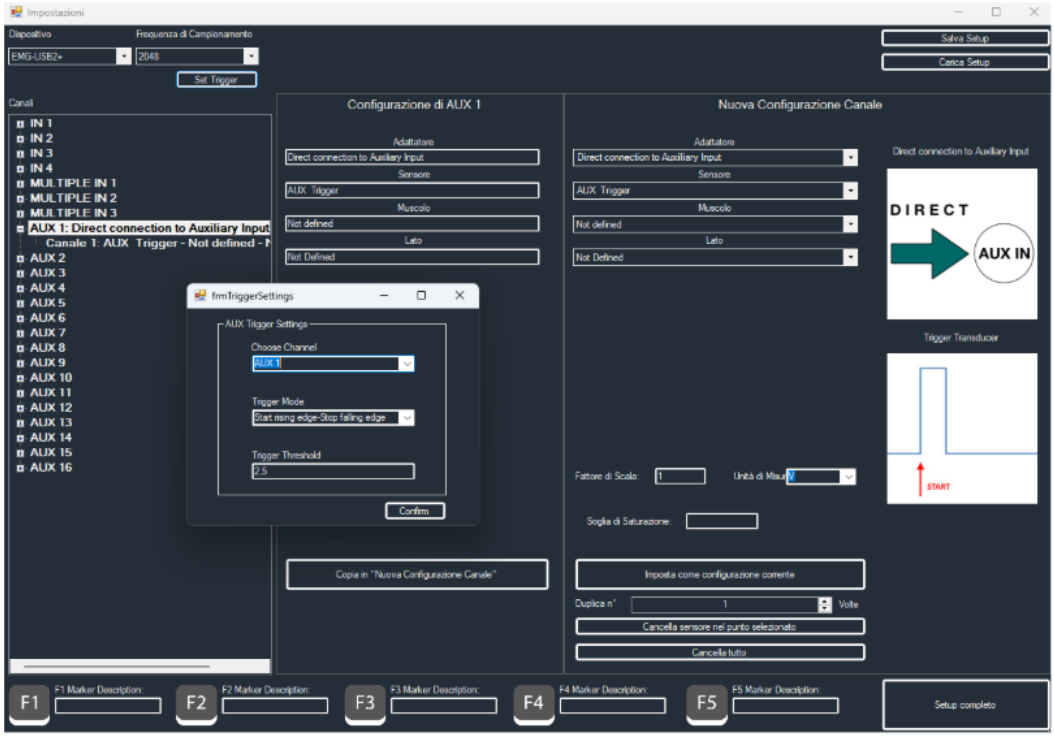


Figure 3.1 Selection of the direct connection to auxiliary input within the OTBioLab+ environment to enable external TTL hardware triggering.

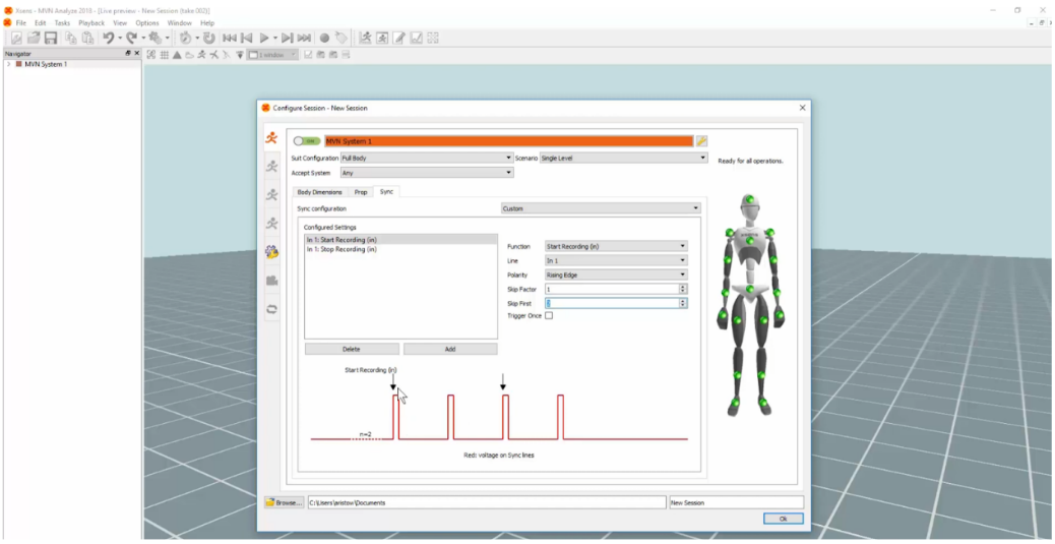


Figure 3.2 Hardware interface and synchronization pulse generation settings within the Movella Xsens software environment.

SEEDS dataset presented significant variability in total duration and in the temporal occurrence of the isometric phases. An initial analysis was conducted to identify the optimal decomposition window for each subject and trial, attempting to extract a sufficient number of reliable motor units. Also, processing the SEEDS dataset required the development of a custom grid organization geometry, as suggested by the MUedit documentation for handling new or unsupported electrode layouts. So, the dataset provided by Guo et al. was selected as the primary data source, for their grids directly compatible with the default array configurations already available in the MUedit library and their higher inter trial repeatability and kinematic consistency.

Their HD-sEMG signals were acquired using a total of 448 channels distributed over the wrist and forearm regions. The first 256 channels were used to record muscular activity from the wrist region through four 5×13 electrode grids (64 channels per grid, 4 mm inter-electrode distance; GR04MM1305, OT Bioelettronica, Italy). The remaining 192 channels were used to record forearm muscular activity through three 8×8 electrode grids (10 mm inter-electrode distance; GR10MM0808, OT Bioelettronica, Italy). The position of the electrode grids and the anatomical references are reported in the Fig 3.3.

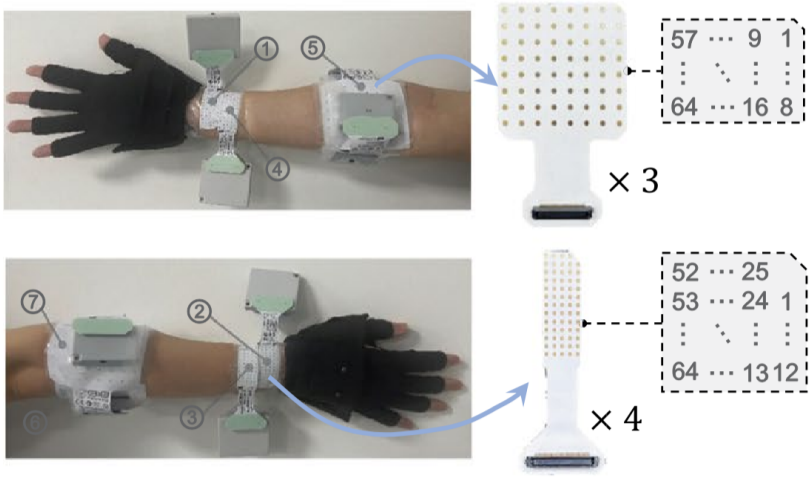


Figure 3.3 Anatomical references and electrode grid placement for the concurrent recording of HD sEMG-signals at the wrist and forearm regions.

The HD-sEMG signals were recorded at a sampling frequency of 2000 Hz with a 16-bit resolution. In parallel, hand kinematics were acquired using a 5DT Data Glove 14 Ultra sensorized glove, sampled at 200 Hz. The glove provides measurements related to finger flexion, extension, and abduction movements through dedicated fiber-optic sensors positioned on the fingers. The synchronized HD-sEMG and glove recordings were collected during the execution of 13 different hand gestures from 10 healthy participants.
For each participant, the experimental protocol consisted of six trials including all 13 gestures. Each gesture involved two dynamic flexion-extension repetitions followed by a 5-second isometric contraction. Consecutive movements were separated by 5-second

resting intervals. The duration of each dynamic contraction was approximately 2 seconds, resulting in an overall trial duration of approximately 187 seconds (Fig 3.4).

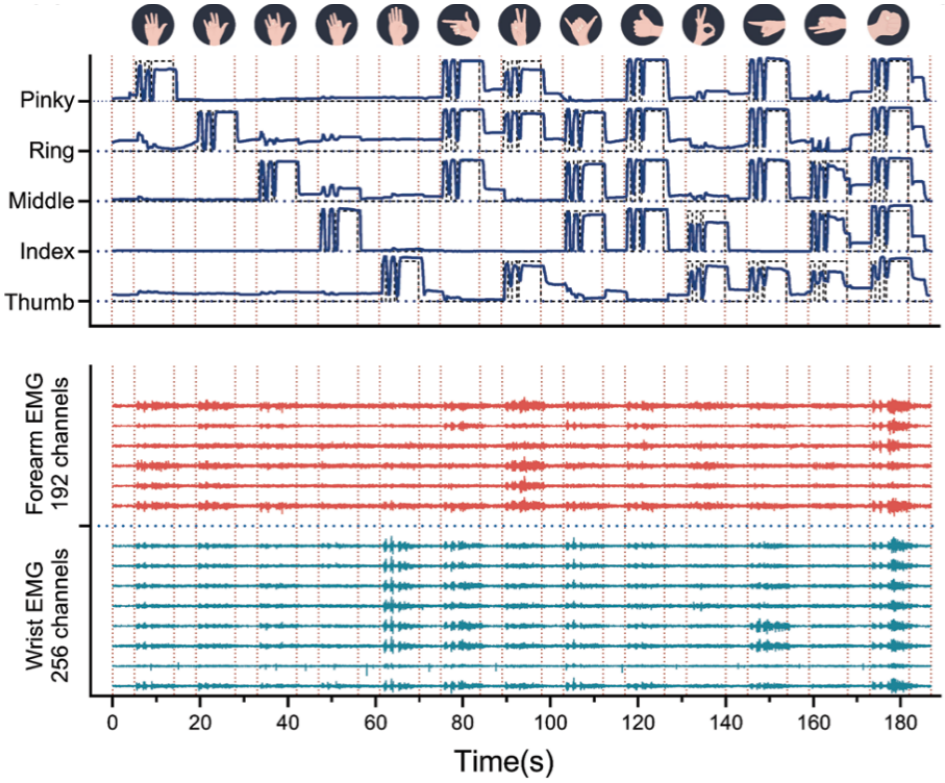


Figure 3.4 Overview of the 13 target movements included in the experimental protocol: 1-Little finger flexion; 2-Ring finger flexion; 3-Middle finger flexion; 4-Index finger flexion; 5-Thumb flexion; 6-Middle-ring-little finger flexion; 7-Thumb-ring-little finger flexion; 8-Index-middle ring finger flexion; 9-Four finger flexion; 10-Thumb and index flexion; 11-Index finger point; 12-Thumb-index-middle finger flexion; 13-Hand closed.

3.2.1 Glove data processing and identification of isometric intervals

The six trials from the first subject are inspected. The HD-FW KIN dataset provides both HD-sEMG and glove recordings in WFDB (WaveForm Database) format. Each recording consists of a header file (.hea), containing information about the acquired signals, and a binary data file (.dat) containing the recorded samples. The signals were imported using the WFDB Python package.

For the glove recordings, the loaded data matrix consisted of 14 channels corresponding to finger joint and inter-finger abduction measurements. The glove signal names and their corresponding anatomical interpretation are reported in the dataset documentation [2]. The resulting matrix has dimensions of 37400 samples × 14 channels for each trial.

[2]"A database of hand kinematics, high-density sEMG of forearm and wrist for motion intent recognition" (Guo et al.) [59].

Although the sampling frequency reported in the WFDB header is 2000 Hz, the glove signals were processed using a sampling frequency of 200 Hz, according to the acquisition specifications reported in the original dataset description.

The raw glove sensor values are initially normalized independently for each channel using min-max normalization, resulting in values within the range [0,1], where the minimum and maximum values correspond to the maximum extension and flexion recorded for each sensor. This normalization has also been done by the authors in the paper for their preprocessing.
The normalized signals are subsequently converted into joint angles expressed in degrees. A channel mapping is applied according to the anatomical function of each sensor. Flexion/Extension channels are converted using the expected range of motion of the corresponding finger joints, while finger abduction channels were centered around zero and scaled according to their expected angular range, as explained in detail in Section 3.4 and visually validated in Unity. The obtained hand trajectories reproduced the expected finger movements with smooth and consistent behaviour. Although some joint angle ranges may require further refinement for a perfect biomechanical calibration, the reconstructed movements are sufficiently accurate for identifying the temporal organization of the gestures and extracting the quasi-isometric intervals required.

The conversion from sensor values to joint angles is performed to obtain a physically interpretable representation of hand motion. In particular, angular velocities expressed in degrees per second allow the definition of a stability criterion based on biomechanical quantities rather than arbitrary sensor units. This representation is therefore preferred for the identification of quasi-isometric phases.

The resulting angular trajectories are used to compute the angular velocity of each joint through discrete differentiation.
Positive and negative velocity values indicate opposite movement directions, corresponding respectively to an increase or decrease of the joint angle (e.g., flexion and extension movements). Since the objective is to identify stable phases independently from movement direction, the absolute value of the angular velocity is considered.

A sample was classified as stable when the absolute angular velocity of all 14 glove channels was below a predefined threshold. The threshold was empirically selected by inspecting the angular velocity profiles and was set to 200°/s.
To avoid the detection of short transient events, only continuous stable intervals with a minimum duration of 3 seconds were retained, corresponding to 600 consecutive samples at the glove sampling frequency.

Using this criterion, in most trials, 27 intervals were detected, for example in the first trial (Fig. 3.5), while for other trials 25 are identified. These intervals corresponded to 14 resting periods and 13 isometric contraction phases. The distinction between rest and isometric contraction intervals was assigned according to their temporal position within the predefined task sequence, since the algorithm itself detects only low velocity conditions.

The identified intervals were further verified by comparing their timing with the experimental protocol reported in the original dataset description, where each gesture included a dynamic phase followed by a 5-second isometric contraction and a resting interval. Additionally, the corresponding HD-sEMG signals were visually inspected to confirm that the selected isometric windows contained sustained muscle activation consistent with an isometric contraction rather than only resting activity.

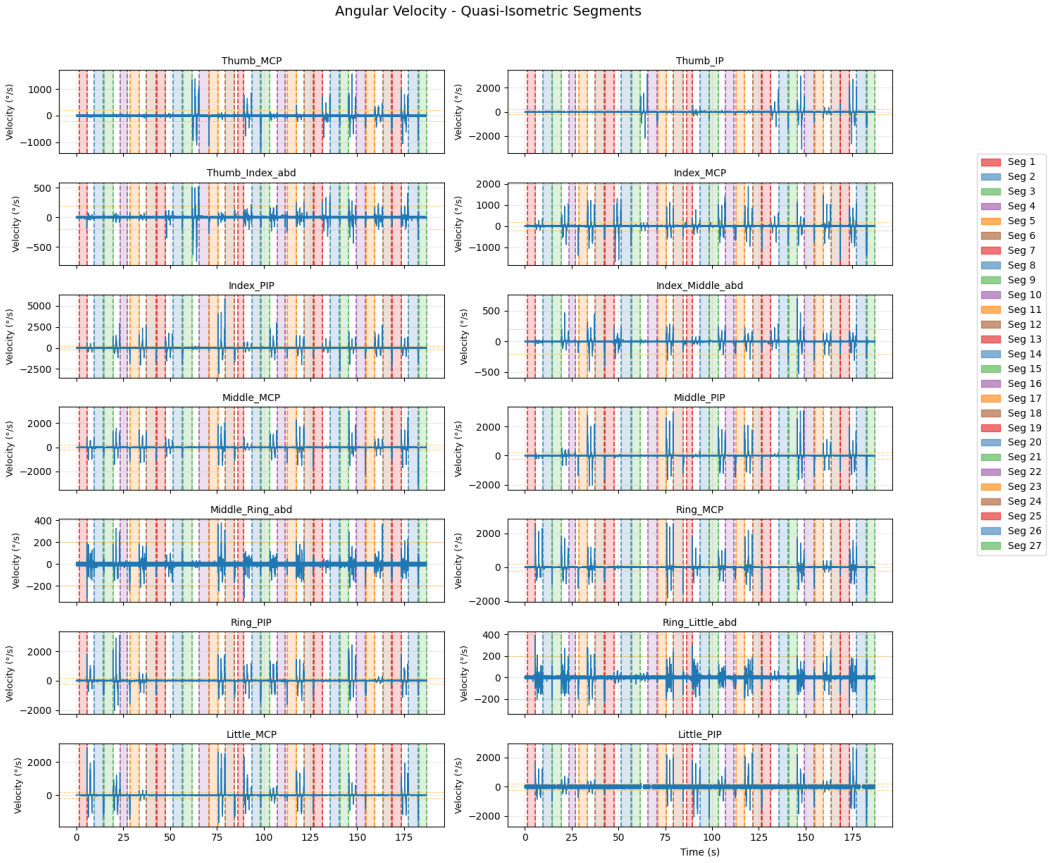


Figure 3.5 Glove angular velocity profiles and identified low velocity intervals corresponding to quasi isometric phases and resting periods.

3.2.2 HD-sEMG extraction and preparation for motor unit decomposition

The isometric intervals identified from the glove recordings were subsequently synchronized with the HD-sEMG recordings in order to extract only the signal portions corresponding to sustained contractions. Since glove and HD-sEMG signals were acquired simultaneously within the same trial, synchronization was performed by converting the temporal boundaries detected from the glove sampling frequency ($f_s = 200$ Hz) into the corresponding HD-sEMG sample indices using the HD-sEMG sampling frequency ($f_s = 2000$ Hz).

The original dataset provides 448 HD-sEMG channels. In this work, only the last 192 channels were considered, corresponding to the three forearm electrode grids used to record the activity of the extrinsic hand muscles. The first 256 channels, associated with the wrist region, were excluded to reduce the complexity of the decomposition process.

Before motor unit decomposition, the extracted HD-sEMG signals were filtered using a fourth order zero phase Butterworth band-pass filter with cutoff frequencies of 20 Hz and 500 Hz, to clean the signal before concatenation, as proposed by Tanzarella et al. (2021).

The selected 192 channels were organized according to the physical electrode configuration of the acquisition system, consisting of three independent 8×8 electrode grids (64 channels per grid). This organization was preserved in the conversion to the MATLAB structure required by MUedit (Fig.3.6).

```
elseif contains(gridname{i}, 'GR10MM0808') || contains(gridname{i}, 'HD10MM0808')
    ElChannelMap = ([8 16 24 32 40 48 56 64; ...
                     7 15 23 31 39 47 55 63; ...
                     6 14 22 30 38 46 54 62; ...
                     5 13 21 29 37 45 53 61; ...
                     4 12 20 28 36 44 52 60; ...
                     3 11 19 27 35 43 51 59; ...
                     2 10 18 26 34 42 50 58; ...
                     1 9 17 25 33 41 49 57]);

    if contains(gridname{i}, 'HD')
        ElChannelMap = fliplr(ElChannelMap);
    end
    discardChannelsVec{i} = zeros(64,1);
    nbelectrodes = 64;
    IED(i) = 10;
    emgtype(i) = 1;
```

Figure 3.6 Array channel configuration and grid mapping for the 8×8 electrode grids within the MUedit library.

3.2.3 Multi-trial Multi-task Decomposition

Following the multi-task decomposition approach proposed in previous HD-sEMG studies (Tanzarella et al., 2021), an initial analysis was performed by investigating, for a single subject, the possibility of decomposing the concatenation of multiple isometric contractions belonging to different tasks in order to track common MU across them. Moreover, unlike Tanzarella et al., also a multi-trial strategy is adopted to increase the total duration of the signal available for motor unit decomposition while maintaining the same subject, electrode configuration, and recording conditions.
Increasing the number of available discharges improves the robustness of motor unit identification, allowing more reliable estimation of MUAP signature and facilitating the detection of motor units consistently activated across different task conditions. Furthermore, it's interesting to analyze whether the use of repeated recordings of the same task allows the assessment of decomposition repeatability and the consistency of identified motor unit characteristics across separate contractions.

Different combination of concatenation were evaluated, including the combination of all 13 available hand gesture or more than 3 trial per task. However, these configurations did not provide a sufficient number of reliable motor units for subsequent analyses. The extracted motor units frequently failed to satisfy the quality criteria required for acceptance, mainly due to the increased variability introduced by combining multiple different tasks and/or trials. This lack of consistency was exacerbated by significant inter-session physiological variability. Since trials for the same task were recorded across different sessions, factors such as muscular fatigue, variations in electrode-skin impedance, and fluctuations in the subject's neurophysiological state introduced non-stationarities that hindered the robust tracking of motor units over time.

Therefore, a more constrained multi-task multi-trial approach was adopted, selecting only two related tasks and two repetitions for each task. The final concatenated signal (Fig.3.7) was composed of four isometric contractions recorded from the same participant (subject 01):

- 5 s precision pinch contraction from trial 1;
- 5 s precision pinch contraction from trial 2;
- 5 s hand close contraction from trial 1;
- 5 s hand close contraction from trial 2.

The resulting signal duration of each task is 10 seconds, which provides a larger number of motor unit discharge events compared with a single 5-second contraction and it is comparable with the literature length which is between 10 and 20 seconds.

The selection of these two tasks was motivated by previous findings reported by Oßwald et al. (2025 [60]), where both task-specific and shared motor units were identified during different single and multi digit hand tasks (Fig.3.8).

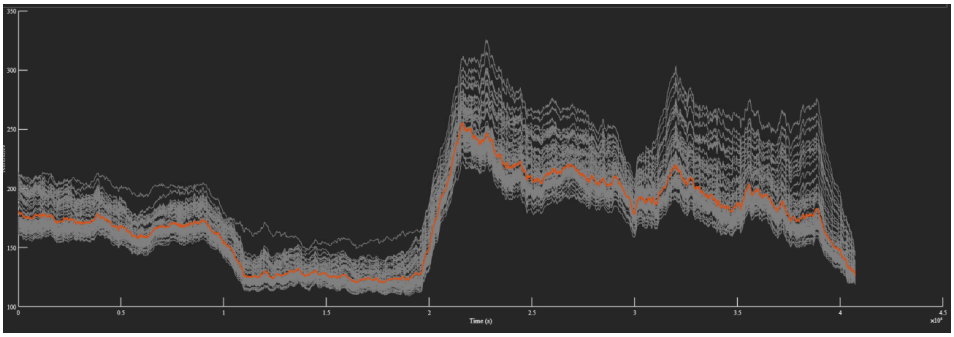


Figure 3.7 Smoothed multi channel sEMG envelope showing the four concatenated contraction intervals selected for the multi task multi trial decomposition pipeline.

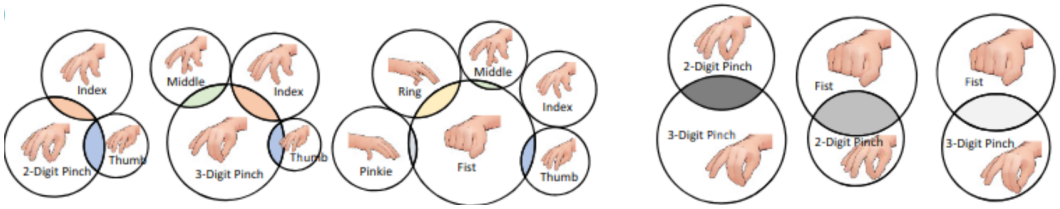


Figure 3.8 Venn diagrams showing the distribution of task specific and shared motor units across single digit and multidigit tasks as reported by Oßwald et al. (2025).

MUedit parameters

Since the HD-sEMG acquisition consisted of three independent electrode grids, the decomposition was performed separately for each grid of 64 channels. This approach preserves the spatial organization of the electrode arrays and allows the estimation of motor unit filters specific to each anatomical recording region.

The parameters used for this experiment were selected according to commonly adopted criteria in HD-sEMG decomposition studies, with adaptations required by the characteristics of the present dataset, including short isometric contractions and the concatenation of different motor tasks and trials.

The main parameters used for MU identification and validation are summarized below (Fig.3.9).

- **SIL threshold**: in literature, a strict SIL threshold above 0.90 is commonly adopted to ensure high physiological reliability, especially for long and stable isometric contractions performed at constant force. In the present study, the threshold was set to 0.85. The stricter threshold of 0.90 resulted in the rejection of almost all extracted motor units, mainly due to the shorter duration of the available contractions and the presence of additional variability introduced by concatenating different task conditions.

- **Coefficient of variation**: is excluded as a strict rejection criterion. This parameter evaluates the regularity of motor unit discharge frequency. Although values below approximately 30% are generally considered indicative of stable physiological discharge patterns, this criterion becomes less suitable for the present dataset. A motor unit may be strongly recruited during one task (e.g., precision pinch) and partially or completely silent during another task (e.g., hand close) and the same could be for different trial of the same task.

- **Extension factor**: an extension factor equal to 4 was selected. This parameter increases the number of observations available for source separation by creating delayed versions of the original EMG channels. Therefore, each 64-channel grid was expanded to 256 extended channels. This approach is based on the physical propagation of the muscle action potential: since a single MU discharge travels along the muscle fibers at a finite conduction velocity, it is detected by adjacent electrodes at different times. By including temporal delays, the algorithm can capture the same MU activation at different stages of its propagation across the grid. This increased dimensionality essentially allows the separation algorithm to leverage both spatial information (from the

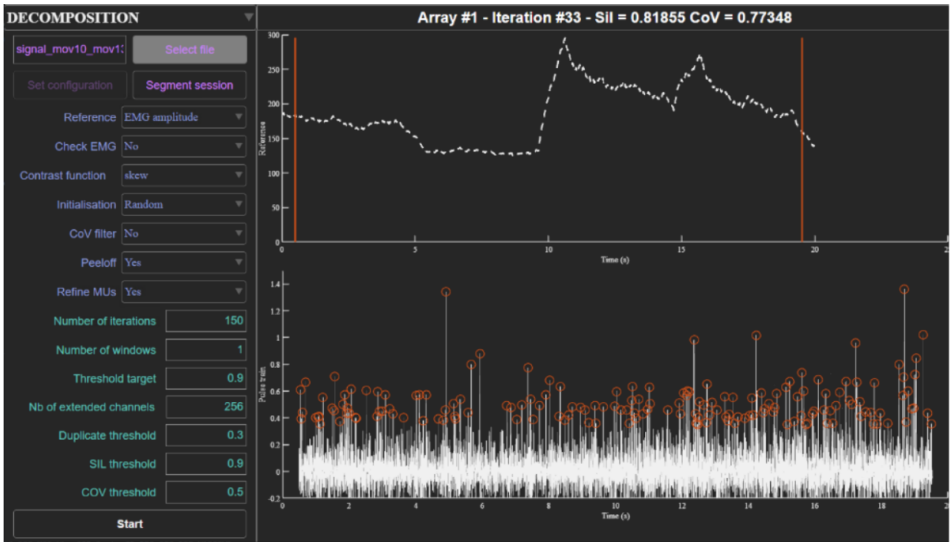


Figure 3.9 MUedit Decomposition Interface on Matlab

grid layout) and temporal information (the conduction delay), significantly improving the estimation of the spatial filters and the isolation of individual neural sources.

- **Duplicate threshold**: motor units with highly overlapping discharge patterns were considered duplicates. A threshold of 30% common discharge events was used to identify and remove redundant motor unit detections.
- **Peel-off**: the peel-off procedure was enabled to iteratively remove the contribution of already identified reliable motor units from the residual EMG signal. This allows subsequent decomposition steps to focus on remaining unresolved sources.
- **Motor unit refinement**: the refinement procedure was enabled to improve the temporal consistency of the extracted spike trains. The algorithm iteratively removes abnormal discharge events characterized by instantaneous firing rates exceeding three standard deviations from the mean firing behaviour. After removal of inconsistent discharges, the separation filter is recalculated using the refined spike train and subsequently applied to the continuous EMG signal.

To further discriminate true neural discharges from noise or silent periods associated with task transitions, the extracted peak amplitudes, in MUedit, are classified using a K-means clustering algorithm, allowing the selection of the cluster corresponding to physiologically consistent motor unit action potentials.

MU quality criteria

Motor unit decomposition algorithms provide a set of candidate motor units. However, additional validation steps are required to distinguish physiologically meaningful motor units from unstable decompositions, false detections, or duplicated sources.

Different approaches have been proposed in the literature depending on the experimental paradigm.

For long isometric contractions, Tanzarella et al. (2021), after the convolution kernel compensation (CKC) decomposition, applied a quality control procedure based on the PNR, removal of false discharge peaks through manual visual inspection, and a minimum number of discharges within each analysed contraction. Motor units with PNR values below 25 dB were discarded, and units were considered active during a specific grip only if they generated at least 20 discharges within the analysed 10 seconds interval. To verify whether the same motor unit was identified across different grip conditions, they decomposed the concatenated signal and compared the estimated MUAP waveforms through two-dimensional cross-correlation of the spike triggered averaged potentials. The obtained correlation values provide a quantitative reference for evaluating MU consistency: motor units identified as the same physiological source across all grip conditions showed a mean two dimensional cross-correlation coefficient of 0.91, whereas comparisons between different motor units resulted in a lower mean correlation value of 0.64.

In addition to MUAP similarity analysis, Tanzarella et al. reported the number of identified motor units for each participant and muscle across the seven analysed movements (Fig. 3.10). These values provide a reference range for the expected number of physiologically plausible motor units extracted from HD-sEMG recordings. An example of the resulting motor unit discharge trains reported by Tanzarella et al. is shown in Fig. (Fig. 3.11), illustrating the temporal activation patterns expected after a successful decomposition.

Furthermore, Tanzarella et al. divided the electrode grids into predefined channel bands in order to associate the identified motor units with specific anatomical muscle regions. In the present study, however, the motor units extracted from the three HD-sEMG grids positioned over the forearm were treated as a single neural pool, without applying a priori anatomical segmentation of the channels into different muscle compartments. This choice was motivated by the objective of the subsequent analysis, where the extracted motor unit activity was used for muscle synergy identification rather than anatomical motor unit classification.

Conversely, Oßwald et al. (2025) investigated dynamic single and multi digit tasks by decomposing each task independently. Here, additional inclusion criteria required a mean firing rate (FR) between 5-40 Hz to avoid fast and so probably noisy spikes.
Motor unit matching across tasks was performed through MUAP spatial temporal comparison. The waveform similarity was evaluated by selecting the channels with the highest peak-to-peak MUAP amplitude and calculating normalized cross-correlations between corresponding channels. A correlation value above 0.90 was used as a criterion for identifying the same motor unit across different tasks.

Happold et al. (2026, [61]), using MUedit for motor unit decomposition, adopted a similar post hoc tracking strategy. In particular, they concatenated spike trains fort

Table 1. Number of identified motor neurons associated with each muscle across the total 379 identified motor neurons

Number of identified motor neurons for each muscle									
Muscles	s1	s2	s3	s4	s5	s6	s7	Mean	Std
FDI	12	11	6	13	8	6	13	9.9	3.1
II DI	6	6	3	7	5	1	8	5.1	2.4
III DI	6	5	2	1	1	6	3	3.4	2.2
IV DI	2	3	4	2	0	0	2	1.9	1.5
ADM	10	6	1	6	9	5	11	6.9	3.4
FPB	0	0	0	2	0	0	1	0.4	0.8
APB	4	2	0	0	4	3	4	2.4	1.8
OPP	2	2	1	8	5	5	4	3.9	2.4
ECU	4	5	2	2	3	4	6	3.7	1.5
EDC	2	5	6	8	5	2	0	4.0	2.8
ECR	2	1	2	0	1	2	3	1.6	1.0
FCU	4	3	4	6	3	2	2	3.4	1.4
FDP	4	4	0	4	7	2	9	4.3	3.0
FCR	8	2	4	3	1	2	3	3.3	2.3

Values for each subject and also mean and SD across subjects are reported.

Figure 3.10 Distribution and average number of identified motor units per participant across different group muscles as reported by Tanzarella et al. (2021).

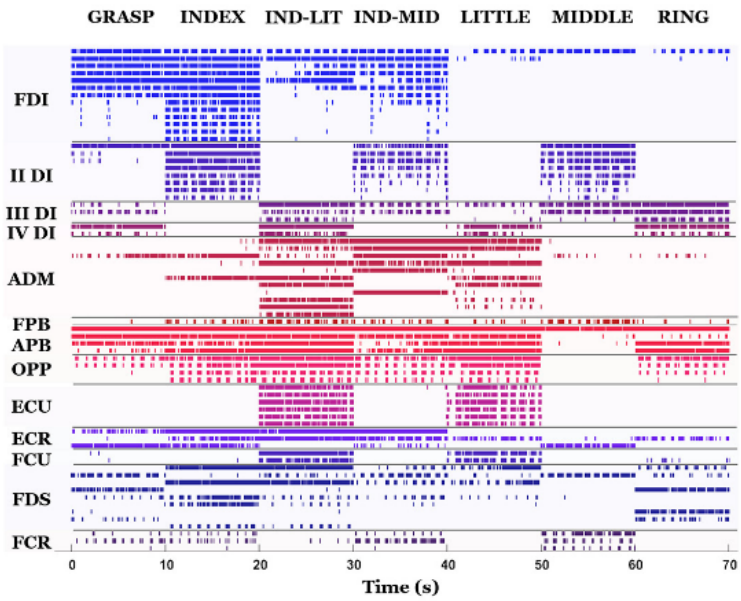


Figure 3.11 Example of motor unit discharge trains and characteristic temporal firing patterns following CKC decomposition from Tanzarella et al. (2021) study.

the same MU tracked across 6 target movements to compute motoneuron synergies. They also found a mean of 12 MU (on two grids) per movement.

So, in this work the MU obtained by FASTICA in MUedit are processed in two steps:

- The global discharge trains obtained from the concatenated signal were separated into their corresponding temporal segments, each representing a single 5 second isometric contraction. For each segment, false positive detections were removed, with a maximum physiologically plausible instantaneous discharge frequency set to 40 Hz.
  To determine whether a motor unit represented a true biological activation rather than a sporadic noise component, a minimum discharge criterion was defined. Happold et al. (2026) considered a motor unit reliable when at least 30 spikes were detected during a continuous 10 second movement. Since the present dataset consisted of independent 5 second isometric contractions, the equivalent minimum requirement was scaled to 15 spikes per segment.
  Because task related motor units may not necessarily be active during every repetition of the same movement, the criterion was not imposed simultaneously on both repetitions. A motor unit was retained if it exceeded the minimum discharge threshold in at least one of the two trials belonging to the same task.

- The STA technique is used to estimate the spatio temporal profile of each MUAP. Considering the continuous EMG signal recorded from a given channel and the set of discrete discharge times (expressed in samples) detected for a specific motor unit, the MUAP waveform was estimated by averaging the EMG segments extracted around each detected discharge event (Eq. 3.1):

$$MUAP_c(\tau) = \frac{1}{N_s} \sum_{k=1}^{N_s} EMG_c(t_k + \tau) \tag{3.1}$$

  where τ represents the temporal offset relative to the spike trigger. A temporal window ranging from -10 ms to +10 ms around each discharge was selected for MUAP extraction, as reported by Martinez-Valdes et al. (2017, [62]) and subsequently referenced by Tanzarella et al. (2025). Given the sampling frequency of 2000 Hz, each estimated MUAP waveform consisted of 50 samples around the discharge event (with the central spike corresponding to the trigger sample).

  Following the approach proposed by Tanzarella et al., the similarity between MUAPs was evaluated through two dimensional normalized cross-correlation. Unlike a conventional one dimensional waveform comparison, this approach considers the MUAP as a spatial temporal matrix, where rows represent the HD-sEMG channels and columns represent the temporal samples of the action potential waveform.

  The normalized cross-correlation evaluates the similarity between two MUAP matrices by testing all possible spatial temporal alignments and selecting the

maximum correlation value. The normalization compensates for differences in signal amplitude between recordings by removing local mean values and scaling by the local standard deviation, allowing comparison of waveform morphology independently from absolute amplitude variations. The resulting correlation coefficient ranges from -1 to 1, where values close to 1 indicate high spatial temporal similarity (Eq. 3.2):

$$R(u,v) = \frac{\sum_{x,y}[A(x,y) - \bar{A}_{u,v}][B(x-u,y-v) - \bar{B}_{u,v}]}{\sqrt{\sum_{x,y}[A(x,y) - \bar{A}_{u,v}]^2 \sum_{x,y}[B(x-u,y-v) - \bar{B}_{u,v}]^2}} \tag{3.2}$$

where $A(x,y)$ represents the reference MUAP matrix, while $B(x-u,y-v)$ represents the second MUAP matrix shifted by a spatial temporal displacement (u,v). The terms $\bar{A}_{u,v}$ and $\bar{B}_{u,v}$ indicate the local mean values of the two matrices at the considered alignment position. The coefficient $R(u,v)$ represents the normalized correlation value obtained for the specific spatial temporal shift.

In this analysis, MUAP consistency was evaluated at different levels. First, motor units identified across repeated trials of the same task were compared to assess inter trial stability. Second, for motor units detected in different tasks, MUAP similarity was evaluated to determine whether the same neural source maintained a consistent spatial signature despite changes in motor task. For shared motor units detected in multiple segments, comparisons were performed using the average MUAP waveform obtained within each task condition. Comparisons between different tasks required at least 30 spikes accumulated within each task, allowing a more robust estimation of the average MUAP waveform. Motor units detected in only one of the two repetitions were not discarded at this stage, since the absence of activity in a single trial may reflect physiological variability in recruitment rather than an unstable decomposition.

Finally, the same MUAP cross-correlation analysis was applied between different extracted motor units to identify possible duplicate detections generated by the decomposition algorithm. Two motor units were considered potential duplicates when their MUAP similarity exceeded the predefined correlation threshold across the available segments. In case of duplicate identification, the motor unit with lower MUAP amplitude and therefore lower signal reliability was discarded.

3.2.4 Trial Upsampling and Normalization Strategy

Although the adopted inclusion criteria ensured a minimum number of discharges per motor unit segment, the relatively short duration of each contraction (5 s) may still lead to limited statistical robustness for MUAP estimation and discharge validation. In addition, inter trial comparisons revealed residual variability in MUAP morphology within the same task, despite high spatial temporal correlation values, suggesting the presence of physiological variability and sensitivity of the estimation

procedure to low spike counts.

To address these limitations, two complementary preprocessing strategies were considered.

The first approach consists of amplitude normalization at the single trial level. Specifically, each trial was normalized using its mean RMS value computed across all channels and rescaled to the range [0, 1] through min–max normalization, with the goal of reducing inter-trial amplitude variability and improving the comparability of MUAP waveforms across repetitions of the same task.

Within this framework, a subset of trials was selected based on visual inspection of signal stability across the available recordings of the first subject. In particular, among the 6 trials available for the first subject, trials 2 and 6 were identified as the most stable for both pinch and hand closed tasks (Fig. 3.12). Importantly, the same trial indices were retained across the two tasks in order to maintain consistency in inter-task comparisons and to avoid introducing additional variability due to unequal trial selection. This decision was motivated by preliminary analyses in which different trial combinations were considered across tasks.

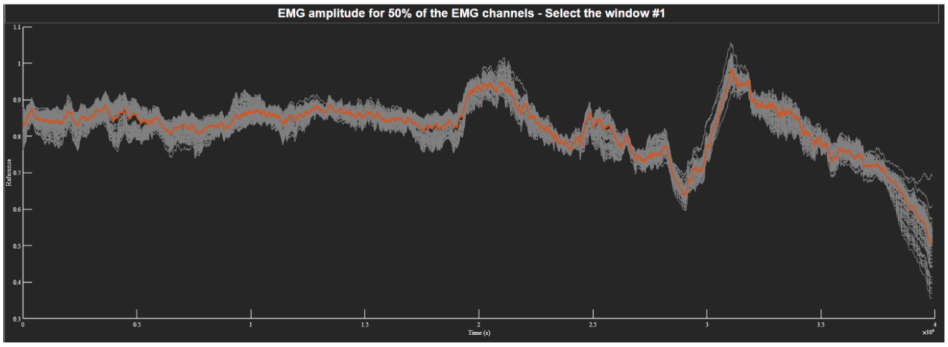


Figure 3.12 Normalized multi channel sEMG profiles for the pinch and hand closed tasks utilizing mean RMS scaling and min max normalization on selected stable trials.

The second approach involves temporal upsampling of the EMG recordings. The original signals sampled at 2000 Hz were interpolated to 4000 Hz prior to decomposition, effectively increasing the temporal resolution of the continuous EMG signal because the decomposition pipeline is still constrained to the original physiological firing resolution. In this framework, a single contraction task now lasts 10 seconds, improving the temporal precision of MUAP alignment and STA waveform estimation. The chosen trial for both tasks is the second one (Fig. 3.13).

3.2.5 MU filters cross-dataset application

Given the set of validated motor units after the postprocessing quality criteria, the corresponding separation filters were stored to enable their application to new EMG recordings and to assess the stability of previously identified motor unit sources

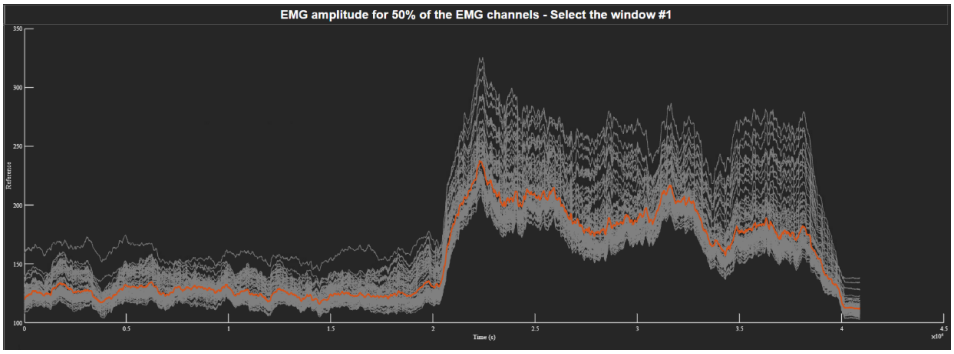


Figure 3.13 Temporal upsampling profiles and interpolated waveforms for the pinch and hand closed contraction segments used to increase alignment resolution.

across datasets.

Within the MUedit pipeline, additional modifications were introduced to preserve all parameters required for reproducible source separation and signal reconstruction. In particular, beyond the estimated pulse trains and discharge times, the decomposition structure was extended to store the whitening transformation matrices and FastICA parameters required to maintain a consistent geometric representation of the signal space.

The multichannel EMG signal is first preprocessed through band-pass and notch filtering to remove artifacts. The signal is then spatially and temporally expanded using a fixed extension factor (exFactor), which concatenates delayed versions of each channel, forming the extended observation matrix. This step increases the dimensionality of the input space and enhances source separability.

Prior to FastICA decomposition, the extended signal is whitened using Zero phase Component Analysis (ZCA). Specifically, the covariance matrix of the extended signal is decomposed into eigenvectors and eigenvalues, yielding the whitening matrices E and D. These matrices are stored together with the decomposition output to ensure that any new target recording can be projected into the same whitened feature space as the original training data.

FastICA is then applied to the whitened signal to estimate the unmixing vectors (MUFilters). The algorithm seeks projection directions that maximize statistical non Gaussianity, exploiting the fact that motor unit spike trains are sparse and non-Gaussian relative to the mixed EMG signal. The optimization iteratively updates each weight vector using a fixed-point scheme, followed by orthogonalization with respect to previously extracted components to avoid redundancy, and normalization to ensure convergence stability. The resulting vectors define spatial filters that correspond to individual motor unit sources.

For application to new datasets, a dedicated script was implemented to reconstruct motor unit activity from previously identified filters. The target EMG signal

is processed using the same preprocessing steps (band pass filtering, extension via exFactor) and then whitened using the stored matrices E and D, ensuring consistency in the feature space.

The saved MUFilters are then applied to the whitened target signal to obtain the estimated source projections. Among the different preprocessing strategies evaluated for improving MU consistency, the upsampling approach was selected for the final reconstruction procedure. Therefore, the validated MU filters were reapplied to the upsampled version of the concatenated pinch and hand closed recordings, providing a cleaner and more consistent set of discharge trains.

The resulting continuous signals are postprocessed to enhance spike detectability and to convert them into discrete discharge trains. Candidate peaks are identified using a minimum inter-peak distance constraint consistent with the physiological refractory period of motor neurons. Detected peaks are then classified based on amplitude using k-means clustering (k = 2), separating good motor unit discharges from residual noise and cross-talk components. The cluster corresponding to higher amplitude spikes is retained as the valid discharge train.
Finally, validated discharge times and pulse trains are stored.

The validated MU filters obtained from the reference trial (the second one) are subsequently applied to all remaining available recordings. This procedure allowed the reconstruction of the discharge activity of the same motor unit population across different trials, providing a consistent neural representation for the subsequent synergy analysis.

3.2.6 Motoneuron Synergies

The discrete discharge times obtained from motor unit decomposition cannot be directly used as input for muscle synergy extraction methods such as Non-Negative Matrix Factorization (NNMF), since they represent instantaneous events analogous to Dirac delta functions rather than continuous neural activation signals. Therefore, the motor unit discharge trains were converted into continuous signals representing the temporal modulation of motor neuron firing rate.

For each motor unit, a binary spike train was generated by creating a vector of zeros with a duration corresponding to the entire analysed trial and assigning a value of one at the time samples corresponding to detected discharge events.

This binary representation was then low pass filtered using a fourth order zero phase Butterworth filter, implemented through the `filtfilt` function, with a cutoff frequency of 2.5 Hz (Capsi-Morales et al., 2025 [10]). The low cutoff frequency was selected to remove the high frequency component associated with individual spikes while preserving the slower modulation of motor unit recruitment and discharge rate.
The SDRs obtained using `filtfilt` with a low-pass Butterworth filter may contain

small negative values due to filtering artifacts. These values were clipped to zero before NNMF to guarantee non-negativity.

The resulting continuous signal represents the SDR of the corresponding motor unit, describing its time-varying neural drive during the task. Each SDR was subsequently normalized between 0 and 1 by dividing the signal by its maximum value, in order to compensate for differences in discharge amplitude between motor units and provide comparable inputs for synergy extraction.

The normalized SDRs obtained from all retained motor units across the three HD-sEMG grids were then concatenated to form the global input matrix **S**. In this representation, each row corresponds to the SDR of a single motor unit, whereas columns represent the temporal samples of the analyzed trial.

Muscle synergies were extracted by applying NNMF to the SDR matrix. NNMF decomposes the original matrix **S** into two lower dimensional non-negative matrices (Eq. 2.5): **W** represents the synergy weight matrix, describing the contribution of each motor unit to each synergy, and **H** represents the temporal activation coefficients, describing the evolution of each synergy over time.

The NNMF decomposition was evaluated according to the procedure reported by Capsi-Morales et al. (2025). The number of extracted synergies k was tested in the range from 2 to 9. For each value of k, the factorization was repeated for 10 independent repetitions using different random initializations of **W** and **H**, generated through different random seeds, in order to reduce the influence of local minima inherent to the NNMF optimization procedure. Each factorization was performed for 100 iterations.

At the end of each set of repetitions, the reconstruction performance was evaluated by calculating the coefficient of determination $\boldsymbol{R^2}$ between the original SDR matrix and the reconstructed matrix obtained from the NNMF model. The resulting $\boldsymbol{R^2}$ values were used to quantify the percentage of variance explained by the extracted synergies and to determine the minimum number of synergies required to achieve an adequate representation of the original motor unit activity.

Cross-trial synergy projection with fixed global W

A key step of the proposed pipeline consists in the generalization of the synergy structure across different recordings.

To enable this trial projection, the other trials reconstructed with the MU filters from the reference trial, are now described by their SDRs matrix with the same stuctural organization as the reference dataset.

An iterative matching procedure was implemented to preserve this geometrical structure. For each motor unit originally included in the reference **W**, its exact corresponding counterpart in the new recording is found (matching the original HD-sEMG grid and unit ID).

If the motor unit is present in the new trial and exceeded the minimum spike activity threshold (25 spikes), its new normalized SDR is computed and placed in the exact

corresponding row of $\mathbf{S}_{\text{new}}$.
Crucially, if a motor unit from the reference trial was not recruited or lost in the new recording, its corresponding row in $\mathbf{S}_{\text{new}}$ was populated entirely with zeros. This strict alignment ensures that the structural dimensionality remains intact and the spatial weights in $\mathbf{W}$ are not misaligned.

Once the synergy model was identified on the reference trial, the spatial synergy matrix W was considered as a fixed neural basis representing the underlying motor unit coordination patterns, and only the temporal activation coefficients were re-estimated for each new recording. This corresponds to solving the following constrained factorization problem:

$$S_{\text{new}} \approx W_{\text{ref}} H_{\text{new}}$$

The estimation of H_{new} was performed using a multiplicative update rule consistent with the original Non-Negative Matrix Factorization (NNMF) optimization problem:

$$\min_{H_{\text{new}} \geq 0} \|S_{\text{new}} - W_{\text{saved}} H_{\text{new}}\|_F^2$$

In the implementation, H_{new} is initialized as a positive matrix:

$$H_{\text{new}}^{(0)} = 0.1 \cdot \mathbf{1}_{k \times n}$$

where $k =$ is the number of synergies and n is the number of temporal samples.

The update of H_{new} is then performed using the multiplicative rule derived from Lee and Seung's NMF algorithm (2000, [63]). This is implemented through the following steps:

$$W^T X \equiv W_{\text{saved}}^T X_{\text{new}}$$

$$W^T W \equiv W_{\text{saved}}^T W_{\text{saved}}$$

and the iterative update:

$$\mathbf{H}_{\text{new}} \leftarrow \mathbf{H}_{\text{new}} \circ \frac{\mathbf{W}^T \mathbf{S}_{\text{new}}}{\mathbf{W}^T \mathbf{W} \mathbf{H}_{\text{new}} + \epsilon}$$

where $\circ$ denotes the Hadamard (element-wise) product, and ϵ (10^{-9}) is a small regularization constant added to the denominator to prevent division by zero. This iterative procedure is repeated for 100 repetitions, ensuring convergence of the temporal activation profiles that minimize the reconstruction error of the new data while strictly adhering to the spatial constraints imposed by the reference trial.

The resulting activation matrices H_{new} are then used to evaluate the ability to learn from this synergistic information the pose of the associated hand gesture.

3.3 Pose Prediction

Before addressing the complex multi-task problem, a preliminary and simplified analysis was carried out to verify that the proposed deep learning framework is able to learn the mapping between the extracted neural synergy activations (matrix H) and the normalized glove sensor signals. For this reason, the initial experiments were restricted to a single task, the hand close movement, corresponding to the second task contained in the concatenated decomposition.

Three experimental configurations were considered for the single task scenario in order to progressively evaluate the generalization capability of the proposed model:

- Intra-subject evaluation: the model was trained using the hand close trials of Subject 2 and tested on a different trial of the same subject.
- Cross-subject evaluation (single-subject training): the model was trained using all available hand close trials from one subject and tested on a new one.
- Cross-subject evaluation (multi-subject training): the model was trained using data from multiple subjects and evaluated on a new subject.

After validating the framework under the single-task condition, the same experimental protocol was repeated using the complete multi-task dataset, including both precision pinch and hand close, to assess the ability of the proposed architecture to learn multiple motor tasks simultaneously.

3.3.1 Data Preprocessing and Sliding Window Approach

The target are the normalized raw sensor values from the glove rather than their conversion into joint angles in order to prevent potential error amplification during the angular transformation phase.

The synchronization between the synergy activation signals and the glove kinematics was performed by aligning the two data streams based on their segmentation length saved during the isometric zones extraction.
In particular, also data glove are upsampled up to 400 Hz, but maintaining its frequency to 200 Hz, in order to double its time. Then, synergy data are downsampled to 200 Hz so to guarantee synchronization with data glove (Fig. 3.14) .
The neural activation signals **H** usually present values with an order of magnitude of 10^{-3} because they represents just temporal information, while the weight of the synergies are described by **W**. Their small magnitude can reduce numerical efficiency during optimization, vanishing gradients, and slow convergence. To address this issue, a constant gain factor of 1000 was applied to the raw **H** matrices.

Following amplification, the synergy activation matrix H was standardized using Z-score normalization. To compute the necessary calibration parameters, the mean and standard deviation were calculated globally on the single subject by aggregating all available trials within the training set. This subject specific normalization ensures

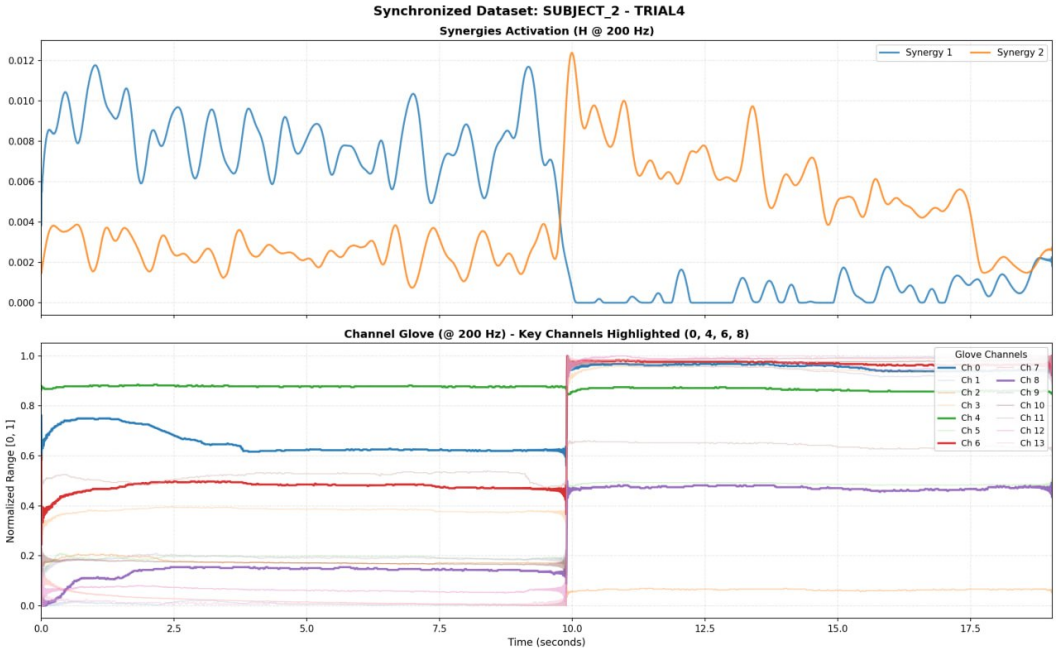


Figure 3.14 Synchronized representation of glove kinematics and high density surface electromyography signals prepared for the training of the recurrent network.

that the physiological baseline and variance of each participant are standardized, mitigating inter subject differences caused by anatomical and recording variations. Crucially, applying a global calibration rather than a trial specific one preserves the relative amplitude variations between different repetitions of the same task. Consequently, the recurrent neural network receives input features that retain both the temporal organization of the synergy activations and the meaningful amplitude modulations related to different contraction intensities, which are essential for accurately scaling the kinematic output.

To map the temporal evolution of the synergies to the continuous kinematics of the data glove, a sliding window approach was implemented. Since the neural and kinematic data were synchronized at a common sampling frequency of 200 Hz, a window size of 40 samples was selected, corresponding precisely to a temporal context of 200 ms. This specific sequence length was chosen to satisfy the strict responsivity requirements for future real time applications, where a system delay greater than 300 ms would significantly degrade the user experience and the fluidity of the bionic control.

3.3.2 Model Architecture

The temporal mapping was performed using a Long Short Term Memory (LSTM) network. The architecture was designed to process the k dimensional input sequence corresponding to the k extracted synergies over the 200 ms window.

The core of the model consists of 2 stacked LSTM layers with a hidden size of 64 units, incorporating a dropout rate of 0.3 to mitigate overfitting on the training trials. The final hidden state of the LSTM is then fed into a fully connected module comprising a hidden linear layer of 32 units with a ReLU activation function, followed by a final linear layer that outputs the 14 degrees of freedom corresponding to the normalized raw values of the glove sensors.

The network was trained for 50 epochs using the Adam optimizer with a learning rate of 0.001.
The loss criterion was defined as the sum of two losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{smoothness}} \tag{3.3}$$

where $\mathcal{L}_{\text{MSE}}$ represents the standard Mean Squared Error used to evaluate the trajectory reconstruction accuracy:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{M \cdot D} \sum_{i=1}^{M} \sum_{j=1}^{D} (y_{i,j} - \hat{y}_{i,j})^2 \tag{3.4}$$

and $\mathcal{L}_{\text{smoothness}}$ is the temporal regularization term (smoothness loss) designed to penalize abrupt velocity variations between consecutive predicted samples:

$$\mathcal{L}_{\text{smoothness}} = \frac{1}{(N-1) \cdot D} \sum_{t=1}^{N-1} \sum_{j=1}^{D} (\hat{y}_{t+1,j} - \hat{y}_{t,j})^2 \tag{3.5}$$

In these formulations, M is the batch size, N is the sequence length (temporal window), D represents the number of kinematic degrees of freedom ($D = 14$), y is the ground truth, $\hat{y}$ is the predicted output, and λ is the regularization weight set to 0.1.

3.4 Unity Visualization

3.4.1 Kinematic Data Preparation

The dataset provides hand kinematics recorded via a data glove defined by 14 channels sampled at 200 Hz. These channels describe finger flexion and inter finger abduction relationships, and the description in the header files specifies the mapping to the following glove sensors:

- **Channel 0 (TN):** Thumb Near flexion
- **Channel 1 (TF):** Thumb Far flexion
- **Channel 2 (T&I):** Thumb Index abduction
- **Channel 3 (IN):** Index Near flexion
- **Channel 4 (IF):** Index Far flexion
- **Channel 5 (I&M):** Index Middle abduction

- **Channel 6 (MN):** Middle Near flexion
- **Channel 7 (MF):** Middle Far flexion
- **Channel 8 (M&R):** Middle Ring abduction
- **Channel 9 (RN):** Ring Near flexion
- **Channel 10 (RF):** Ring Far flexion
- **Channel 11 (R&P):** Ring Pinky abduction
- **Channel 12 (PN):** Pinky Near flexion
- **Channel 13 (PF):** Pinky Far flexion

However, since the public dataset does not provide a standardized 3D rig or a preconfigured mapping strategy, the raw glove values are not directly usable as a plug and play skeletal animation. A custom mapping stage was therefore developed to convert the glove signal representation into local bone rotations for a rigged virtual hand.

Raw glove signals were first normalized independently for each channel using min max normalization over the recorded trial. To drive the 3D visualization, these normalized signals (ranging from 0 to 1) were converted into specific angular ranges suitable for the Unity rig. Flexion channels were linearly mapped to predefined anatomical angular ranges, while abduction channels were centered around zero and scaled to smaller lateral rotation limits. The resulting values were exported as a CSV file to be read directly by the simulation environment. These values are not intended to represent perfectly calibrated anatomical joint angles, but rather a visually plausible retargeting strategy to monitor the network predictions (Fig. 3.15).

3.4.2 Skeletal Rig and Kinematic Retargeting

The 3D right hand model used for the visualization was obtained from the Unity XR package, with the virtual environment developed using Unity Editor version 6000.4.3f1. The glove provides a reduced kinematic description comprising the metacarpophalangeal (MCP) and proximal interphalangeal (PIP) flexion for the four long fingers, the MCP and interphalangeal (IP) flexion for the thumb, and the abduction angles between adjacent fingers.

Conversely, a rigged 3D hand contains more degrees of freedom, including distal interphalangeal (DIP) joints, thumb base orientation, and extra metacarpal helper joints. To bridge this gap, a custom C# script (`HandFingers.cs`) was implemented. The script stores the initial local rotation of each assigned bone. This operation is essential because the imported hand features a native rest pose, and new rotations must be applied relative to this initial state rather than overwriting it entirely.

3.4.3 Empirical Axis Mapping and Joint Approximation

The specific rotation axes were identified empirically by manually manipulating the bones within the Unity Inspector. Finger flexion was mapped to rotations around the local Z axis of each joint, corresponding to the natural closing motion of the fingers. Abduction and adduction were instead applied around the local Y axis exclusively at the level of the MCP joints, avoiding unrealistic twisting of the entire finger chain.

An example of this configuration setup for the little finger components is illustrated in Fig. 3.16. To match the orientation of the Unity skeletal rig, a global sign inversion was required for all flexion related joints, whereas abduction channels were applied without inversion since their directionality was already consistent with the rig coordinate system.

Finally, because the glove does not directly provide DIP angles for the index, middle, ring, and little fingers, a kinematic approximation was introduced to preserve a natural hand shape. The distal joint was estimated directly from the PIP angle using a constant scaling factor:

$$\mathrm{DIP} = 0.6 \times \mathrm{PIP} \tag{3.6}$$

When a third phalanx is available in the rig, the script automatically animates it using this scaled version of the PIP channel, ensuring a fluid and realistic complete finger closure.

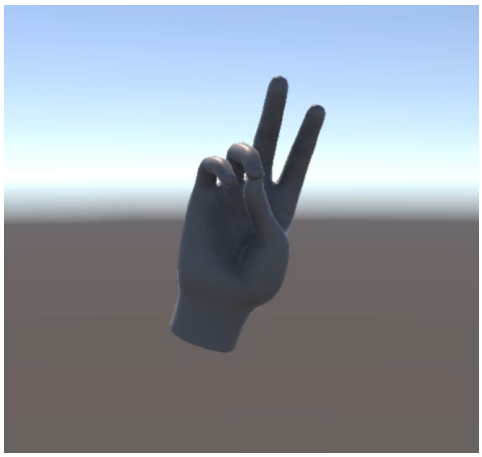


Figure 3.15 Hand pose example extraction from trial 1 of the first subject.

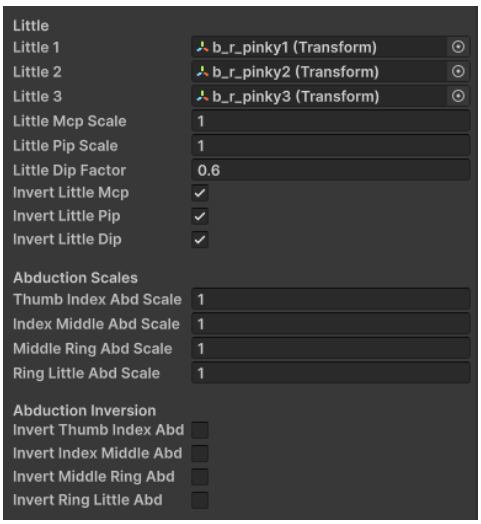


Figure 3.16 Unity pinky transform.

Chapter 4

Results and Discussion

4.1 Motoneuron Synergies

The following decomposition strategy is evaluated on Subject 1 of the HD-FW KIN dataset.
The multi-trial and multi-task decomposition approach was initially considered; however, the analysis highlight relevant variability in MUAP morphology across repetitions, likely related to physiological differences between trials and variations in signal acquisition conditions.

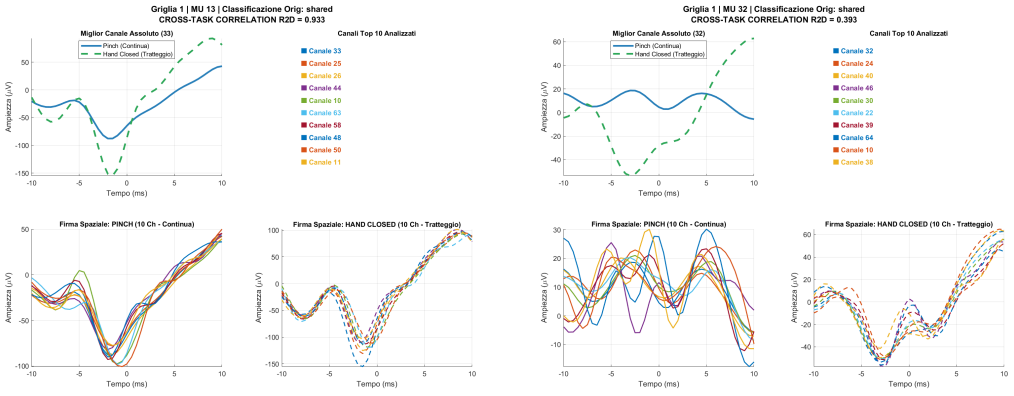
First, the normalization approach was evaluated, consisting of normalizing each trial according to its mean RMS value across channels followed by scaling between 0 and 1. However, this resulted in a less stable decomposition, producing several uncertain motor units that failed the subsequent quality control criteria. In particular, most of the extracted units did not satisfy the required MUAP spatial temporal cross-correlation threshold or the minimum discharge number criterion.
Therefore, this normalization strategy was discarded and the temporal dilation approach was selected as the final strategy to augment the temporal representation of the signal while preserving the original amplitude characteristics.

The motor units retained after the quality control procedure are reported in Fig.4.1, including their corresponding identification number and HD-sEMG grid location. Motor units detected in both pinch and hand closed contractions were considered consistent when their normalized 2D MUAP cross-correlation exceeded 0.7. Conversely, motor units with MUAP similarity values above 0.8 were considered potential duplicate detections and were removed. Only one motor unit was identified as exclusively active during the pinch condition.

Examples of the MUAPs obtained from selected motor units are shown in Fig.4.2. The spatial distribution of the MUAP amplitude highlights the channels mainly contributing to the detection of each motor unit. The corresponding discharge trains separated by HD-sEMG grid are reported in Fig.4.3, showing the temporal activation patterns obtained after the final MUs selection.

```
================================================================
              ELENCO ID DELLE 17 MOTOR UNITS ACCETTATE
================================================================
▶ GRIGLIA 1 (8 unità accettate):
   - MU ID originale:   2  |  Classe Iniziale:   hand_only  |  Stato Finale: Hand_Only_Accepted
   - MU ID originale:  13  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  28  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  29  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  30  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  55  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  85  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  86  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
----------------------------------------------------------------
▶ GRIGLIA 2 (5 unità accettate):
   - MU ID originale:   1  |  Classe Iniziale:   hand_only  |  Stato Finale: Hand_Only_Accepted
   - MU ID originale:   2  |  Classe Iniziale:   hand_only  |  Stato Finale: Hand_Only_Accepted
   - MU ID originale:   8  |  Classe Iniziale:   hand_only  |  Stato Finale: Hand_Only_Accepted
   - MU ID originale:  27  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  28  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
----------------------------------------------------------------
▶ GRIGLIA 3 (4 unità accettate):
   - MU ID originale:   5  |  Classe Iniziale:   hand_only  |  Stato Finale: Hand_Only_Accepted
   - MU ID originale:   7  |  Classe Iniziale:  pinch_only  |  Stato Finale: Pinch_Only_Accepted
   - MU ID originale:  34  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
   - MU ID originale:  39  |  Classe Iniziale:      shared  |  Stato Finale: Shared_Accepted
----------------------------------------------------------------
================================================================
```

Figure 4.1 Summary of identified motor units and their respective grid locations after quality control filtering.



(a) Grid 1 - MU 13

(b) Grid 1 - MU 32

(c) Grid 2 - MU 02

(d) Grid 3 - MU 07

Figure 4.2 Spatial profiles and multi channel waveforms of Motor Unit Action Potentials (MUAPs) for selected representative motor units across different electrode grids.

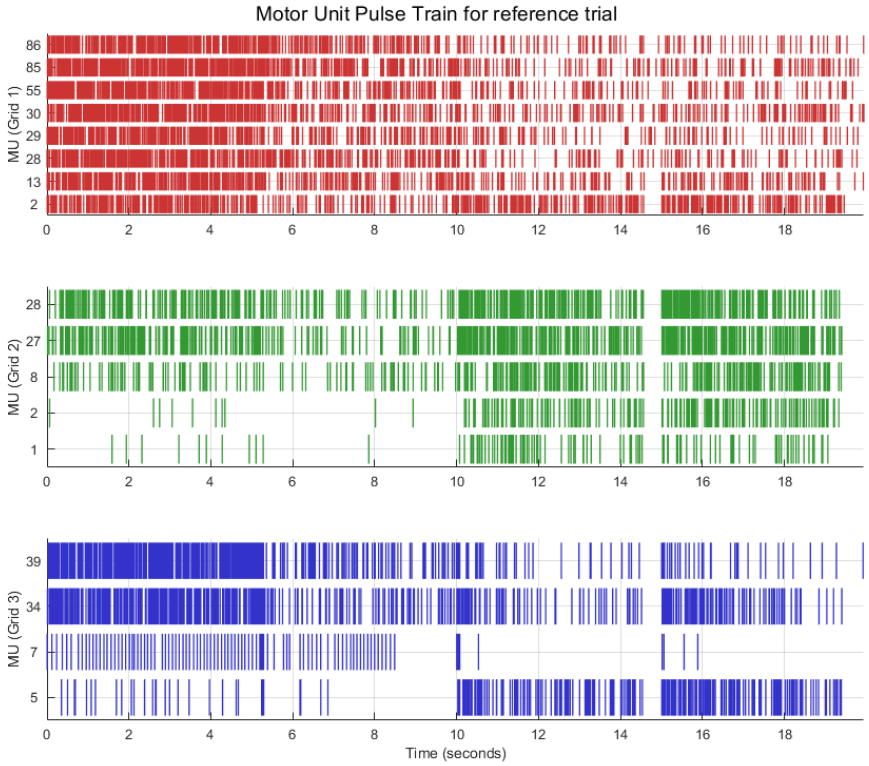


Figure 4.3 Temporal activation patterns and firing behavior of the 17 validated motor units after the final selection process.

For each accepted motor unit, its SDR is computed. An example of SDR estimation for motor unit 2 from grid 1 is shown in Fig.4.4. The temporal profile of the SDR reflects the regularity and modulation of the discharge train, with smoother profiles corresponding to more regular and closer motor unit firing patterns.

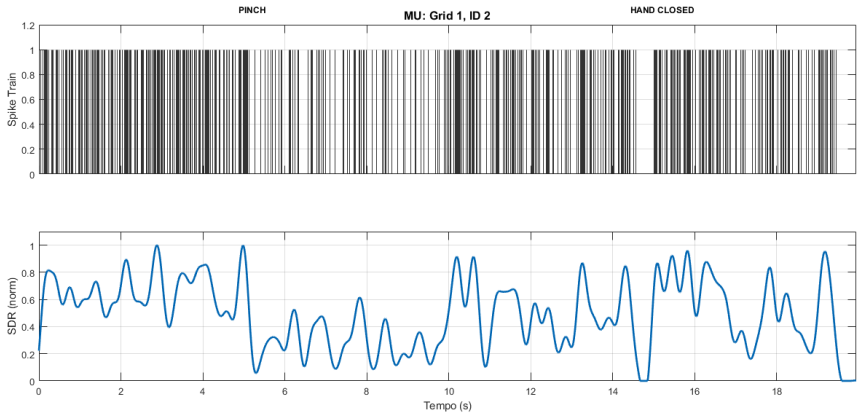


Figure 4.4 Example of the Smoothed Discharge Rate (SDR) estimation over time for motor unit 2 extracted from electrode grid 1.

The validated motor units identified from the reference concatenated recording were reapplied to the remaining available trials using the stored MU filters. Trial 5 was excluded from the subsequent analysis because the hand closed portion of this recording showed insufficient sEMG signal isometry for reliable motor unit reconstruction.

For the synergy analysis, the optimal number of synergies was initially selected according to a reconstruction coefficient threshold of $R^2 > 0.80$, which was achieved with three extracted synergies. However, the third synergy showed limited physiological interpretability, as two synergies exhibited highly similar activation patterns and mainly described overlapping contributions during the pinch movement. Considering both the redundancy observed between these components and the elbow criterion of the reconstruction curve (Fig.4.5), a two synergy solution was selected as the final representation. The two extracted synergies were interpreted as predominantly associated with pinch and hand closed activation patterns, respectively.

The corresponding synergy weighting matrix W and temporal activation matrix H obtained from the reference trial are reported in Fig.4.6 and Fig.4.7. Small high frequency fluctuations observed in the temporal activation profiles H are likely related to the clipping of negative SDR values to zero, which was applied to satisfy the non negativity constraint required by NNMF. The possibility of applying an additional smoothing filter to the estimated activations was considered; however, this was not applied because the subsequent reconstruction using the global W matrix naturally reduced these high frequency components.

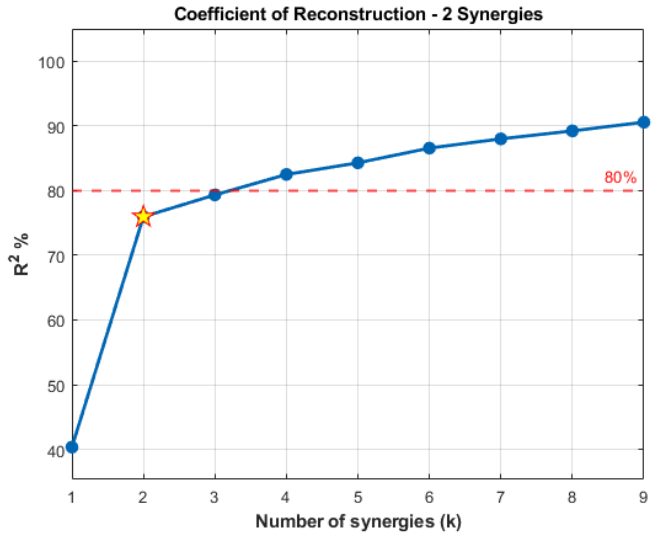


Figure 4.5 Reconstruction coefficient (R^2) as a function of the number of extracted neural synergies, demonstrating the inflection at the chosen two synergy solution.

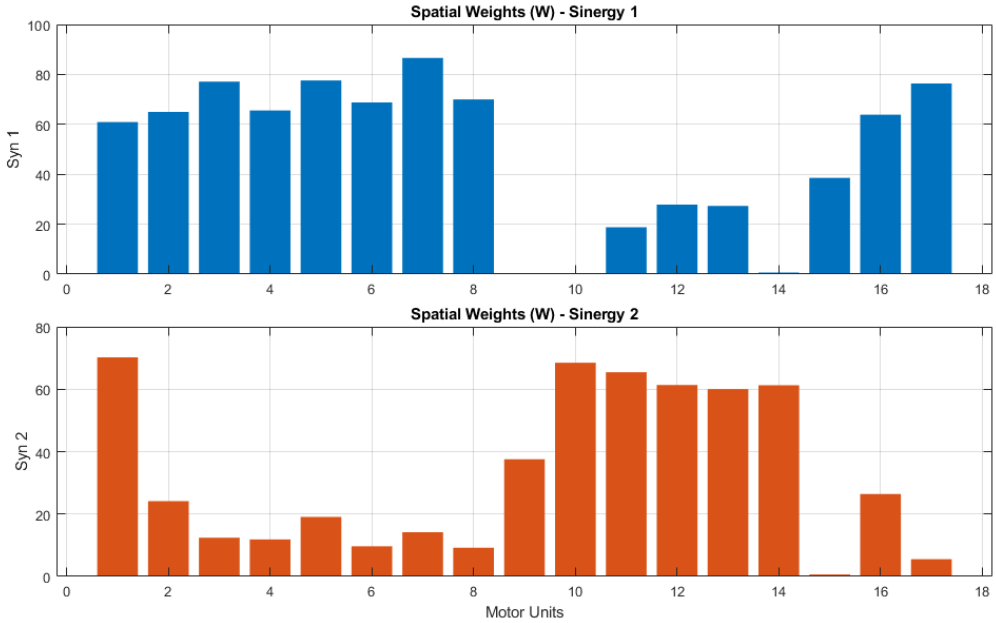


Figure 4.6 Spatial distribution and structural weights of the matrix W representing the muscle synergy groupings for the two extracted control components.

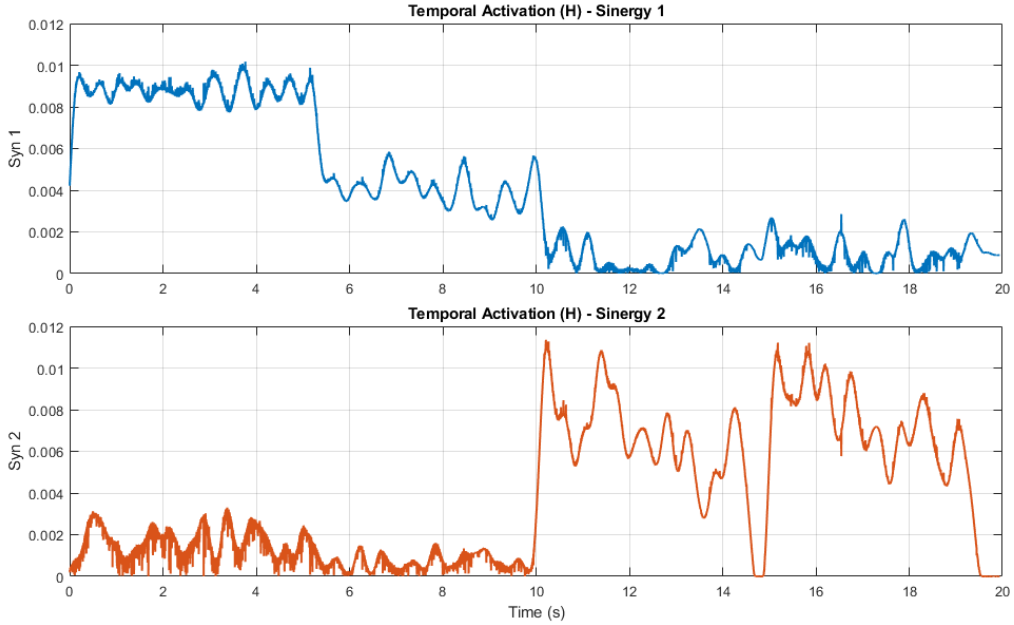


Figure 4.7 Temporal profiles of the synergy activation matrix H during the sequential execution of the target hand tasks in the reference trial.

The accuracy of the H reconstruction is shown in Fig.4.8, where the temporal activations reconstructed using the estimated global W matrix reproduced the original synergy activation profiles of the reference trial, confirming the consistency of the projection procedure.

Finally, the learned synergy structure was applied to the remaining trials using the fixed W matrix (Fig.4.9). Across different recordings, the first synergy, which mainly represented pinch activation in the reference trial, showed a different temporal distribution and contributed to both pinch and hand closed zones. This variability is likely related to the limited number of motor units specifically associated with pinch, where only one validated unit was identified, resulting in higher sensitivity to inter-trial physiological variability.
Conversely, the second synergy, mainly associated with hand closed activation, remained more consistent across trials. This stability is likely due to the larger number of motor units contributing to this movement, providing a more robust representation of the underlying motor command.

4.1.1 Cross subject Analysis

Dataset Extension and Trial Selection

After verifying the feasibility of the proposed methodology, the same processing pipeline was applied to the remaining nine subjects available in the dataset, resulting in a database comprising ten participants. For consistency across subjects, only the Pinch and Hand Closed tasks were considered, as these movements were available

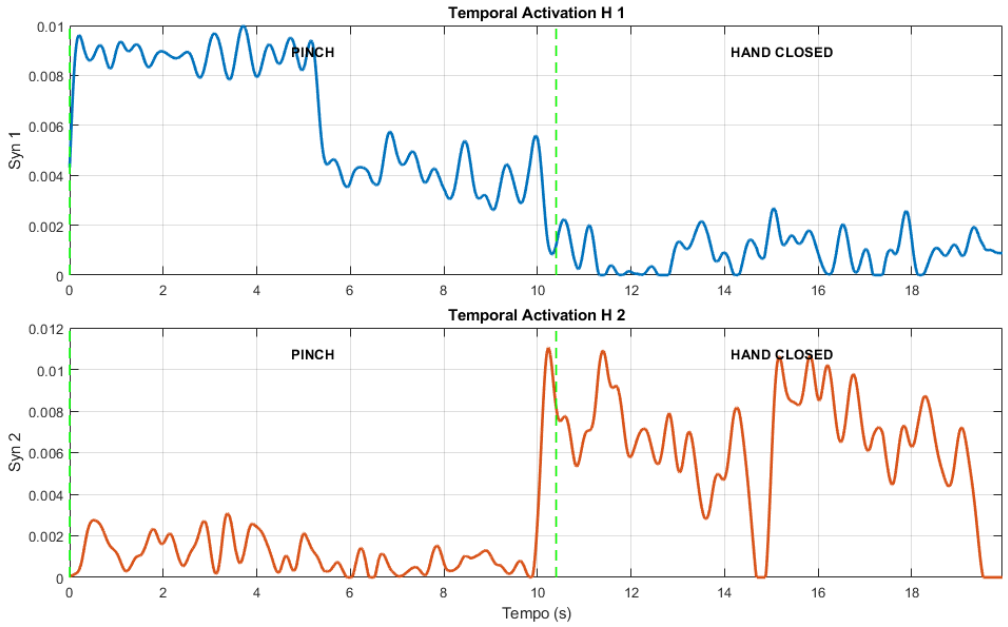


Figure 4.8 Superimposition and verification of the reconstructed temporal profiles against the original synergy activations to validate projection stability.

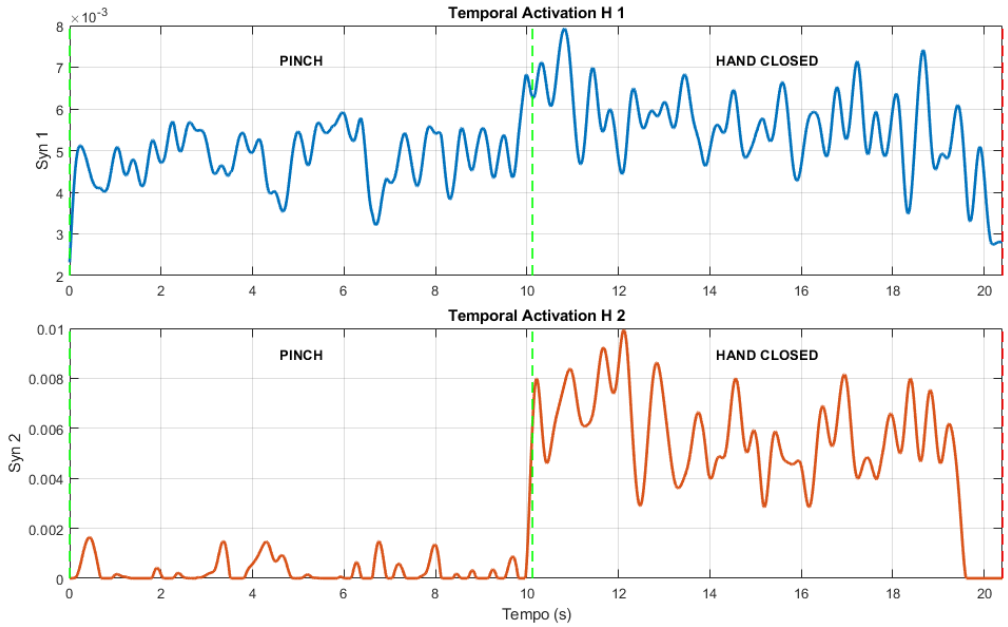


Figure 4.9 Temporal activations of the neural synergies projected across unseen trials using the fixed global weighting matrix. In this case the first synergy represents both taks, maybe due to less MU describing only pinch task in this trial.

for all participants and represented the experimental conditions investigated in this work.

For each subject, to identify the isometric zones for each trial, an angular velocity threshold was empirically selected after visual inspection of the glove signals recordings. Although the same identification procedure was applied to all participants, the optimal threshold varied slightly across subjects, typically ranging between 200°/s and 400°/s. This adjustment was necessary to correctly separate the transition phases from the quasi isometric periods while preserving the expected duration of approximately 5 s for each contraction.

Because some nominally isometric phases still contained residual hand movements, the duration of several identified segments was shorter than expected or exhibited excessive kinematic variability. The trials relative to those segments were excluded from further processing to ensure that only sufficiently stable contractions were used for HD-sEMG decomposition.

Motor Unit Decomposition and Tracking

Among the candidate trials, the reference trial for each subject was chosen as the one providing the most reliable decomposition results, namely the highest number of physiologically plausible motor units together with high quality discharge patterns after criterion application. The identified motor units extracted from the reference trial were then tracked across the remaining valid trials of the same participant to obtain a consistent representation of motor unit activity as done for Subject 1.

Kinematic Variability

During the multi subject analysis, an additional source of variability was observed in the glove recordings. Although the HD-sEMG segmentation was manually verified and the synchronized acquisitions were confirmed to be correct, some glove degrees of freedom exhibited considerable differences across repeated executions of the same task within the same participant. In particular, certain finger flexion channels occasionally showed substantially different amplitudes despite corresponding to nominally identical motor tasks. Consequently, identical neural activation patterns may correspond to different kinematic targets, increasing the uncertainty of the regression problem.

Synergy Extraction and Manual Alignment

Although the reconstruction accuracy generally increases when a larger number of synergies is extracted, the fixed number of two neural synergies is maintained throughout the multi subject analysis. This choice was made to ensure a common latent representation across all participants, thereby simplifying the construction of a unified dataset for the deep learning model.

The ordering of the extracted synergies was manually aligned across subjects to ensure a consistent physiological interpretation. Specifically, whenever the NNMF decomposition produced an inverted ordering of the synergies, they were reordered such that the first synergy predominantly represented the Pinch task (or both tasks

when shared), whereas the second synergy primarily represented the Hand Closed task.

Challenges with Higher Dimensionality

An alternative dataset was also constructed by selecting the modal number of synergies obtained across the different trials of each subject according to an optimal value of the coefficient of reconstruction. The distribution of the selected dimensionalities, together with the corresponding reconstruction accuracy is reported in Fig. 4.10. According to the distribution results, 4 synergies are fixed for each subject.

The alignment of the extracted synergies across subjects represents a non trivial challenge, as mentioned also for the 2 synergies strategy. A possible way to address this issue would involve an additional alignment procedure based on the similarity between synergy activation patterns, for example through cross correlation analysis, in order to identify common latent components across participants. However, in the present dataset, the visual inspection of the extracted components and their activation patterns suggested that the inter subject correspondence was not sufficiently robust to define a reliable common ordering strategy.

The construction of a shared multi subject synergy representation remains an open methodological aspect. This limitation is particularly relevant in learning, since a recurrent neural network such as an LSTM requires input features with a consistent semantic meaning across all samples. Providing the network with input components that represent different neural processes depending on the subject, it would introduce an additional source of variability, potentially preventing the model from learning a meaningful mapping between neural activations and kinematic outputs.

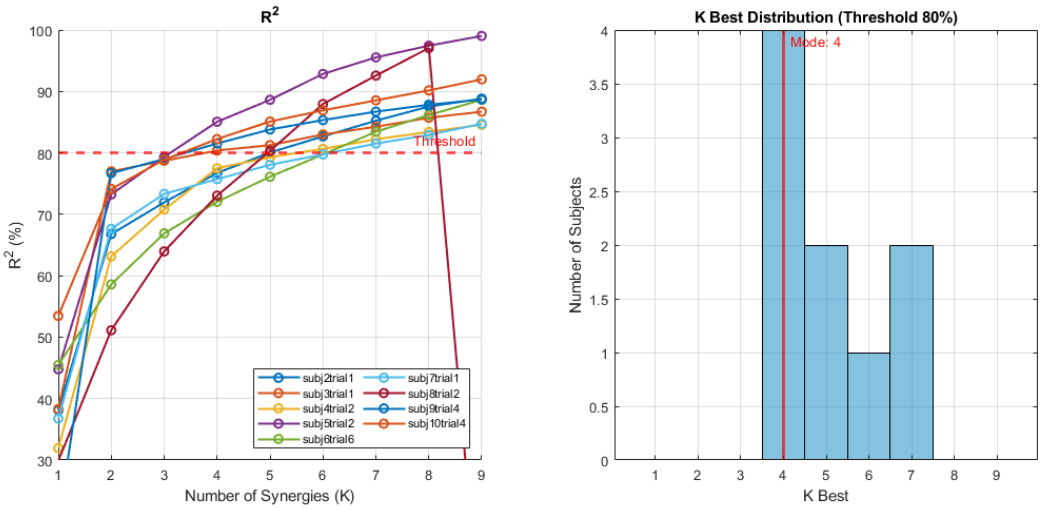


Figure 4.10 Distribution of selected synergy dimensionalities and corresponding reconstruction accuracy.

4.2 Pose Prediction

Before evaluating the performance of the predictive models, a preliminary assessment of the kinematic dataset was conducted to confirm the reliability of the ground truth for the training phase. Visual inspection of the glove sensor channels, further validated through 3D visualizations in Unity, revealed significant inter trial variability among several participants. Subject 2 was specifically selected as the primary reference for the training set because its execution of the movements demonstrated high consistency across all recorded trials (Fig. 4.11).



Figure 4.11 Normalized glove sensor trajectories for Subject 2 across selected channels, highlighting the quasi isometric phases for the Pinch and Hand Closed tasks.

As described in the Methodology, the predictive capabilities of the LSTM were evaluated through a progressive series of experiments, moving from a single task framework to a multi subject concatenated task scenario.

1. **Single Task Evaluation (Hand Closed)**

   To establish a baseline, the model was initially trained to predict a single motor task using a single extracted synergy as input.

   - **Intra-subject validation:** The network was trained on five trials of Subject 2 and tested on the remaining sixth trial. The model successfully learned the prediction mapping almost immediately, confirming the validity of the proposed architecture (Fig. 4.12).

2. **Concatenated Tasks (Pinch and Hand Closed)**

   The complexity was then increased by requiring the network to predict two distinct consecutive movements. For this configuration, the input space was

expanded to two synergies per subject, as determined in the previous synergy analysis chapters.

- **Intra-subject validation:** The network was trained on five trials of Subject 2 and tested on the sixth trial. The model successfully distinguished the transition between the pinch and the full hand closure (Fig. 4.13, Fig. 4.14 (a), (b), (c), (d)).
- **Cross-subject validation:** The model was trained entirely on Subject 2 and tested on all trials of Subject 3 (Fig. 4.14 (e), (f), (g), (h)). As shown in (e) and (g), even tough the ground truth reflects the previous cited problem of the possible variability in the glove recordings, the prediction is tuned on the learned parameters given more precise trials in the training set.

3. **Multi-subject Training for Concatenated Tasks**

   To improve the generalization capability of the model, the training set was expanded to include data from Subjects 2, 7, 8, and 10, while keeping Subject 3 strictly as the test set. Using an input space composed of two synergies, the model successfully predicted the sequence of movements for the unseen subject.
   Expanding the subject pool increased the variability of the ground truth kinematics. Consequently, specific fine motor postures, such as the pinch, resulted in slightly less accurate amplitude predictions compared to the intra-subject model.

   Nevertheless, the network robustly maintained the ability to distinguish between the two macroscopic tasks (Fig. 4.15).

Single Task Evaluation (Training: Subject 2, Test: Trial 4)



Prediction

Figure 4.12 Ground truth and predicted hand kinematics for the Hand Closed task. The network was trained using only the Hand Closed trials from Subject 2 and evaluated on its unseen trial.

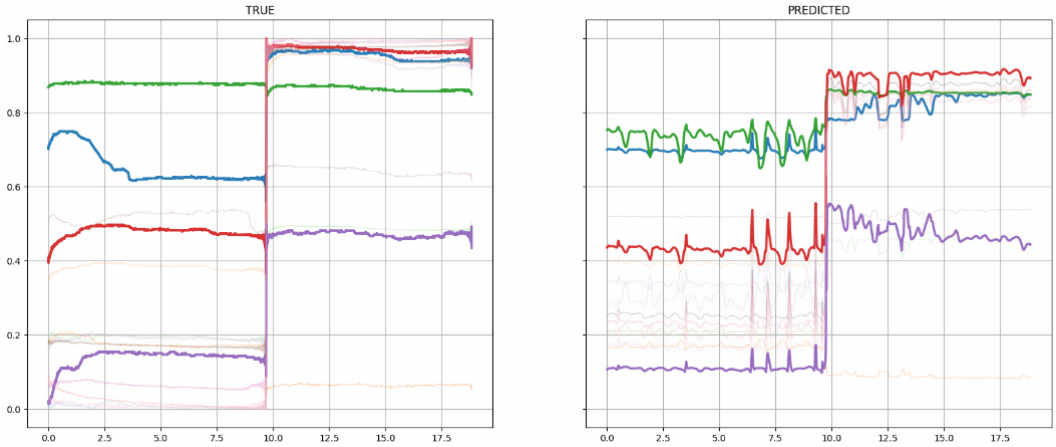


Figure 4.13 Ground truth and predicted hand kinematics for the Hand Closed task on Subject 2 (Trial 4) under intra subject validation.

Synergies Dimensionality and Correlation

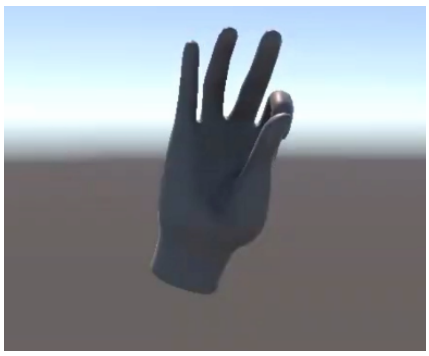
Experimental trials reconstructed by the modal number of synergies, highlighted a fundamental challenge in multi subject learning. When evaluated in an intra subject paradigm, training and testing on different trials of the same participant, the network successfully mapped the 4 dimensional input to the correct hand kinematics, demonstrating that the network has sufficient capacity to handle higher dimensional inputs.

However, when introducing multiple subjects, the model suffered from misclassifications, frequently confusing the pinch and the hand closed phases. This performance drop is not caused by the higher number of synergies, but rather by the lack of physical and semantic alignment of the extracted synergies across different individuals. Because the NNMF extracts synergies independently between subjects, the columns of the input matrix H carry different physiological meanings from one subject to another. Providing the LSTM with unaligned inputs introduces inconsistent features across samples, preventing the recurrent architecture from identifying a stable, shared mapping for the sequential tasks.

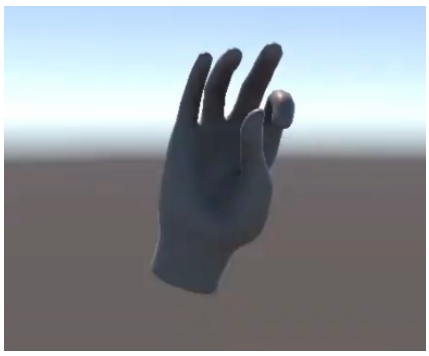
4.3 Future Works

The results obtained in this thesis establish a promising proof of concept for neural driven hand posture control. It is demonstrated that a Long Short Term Memory (LSTM) network can map motor unit synergy activations H to the normalized trajectories of a 14 channel kinematic glove. With a context window of 200 ms, corresponding to 40 samples at a 200 Hz sampling rate, the model proves capable of predicting both single task, given 1 synergy, and concatenated task, with 2 synergies, across both intra subject and inter subject validation paradigms, given an initial calibration for the new subject.
These kinematic predictions can be converted directly into angular degrees of freedom

Intra subject (Subject 2, Trial 4)



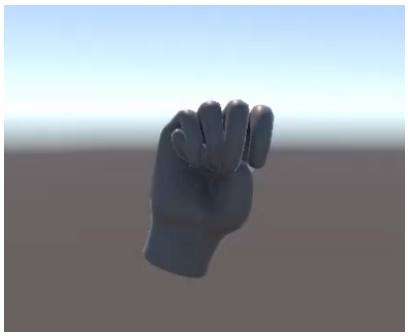
(a) Ground Truth - Pinch



(b) Prediction - Pinch



(c) Ground Truth - Hand Closed

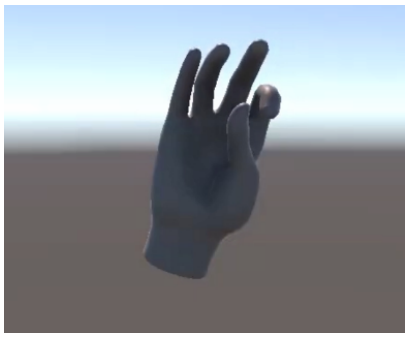


(d) Prediction - Hand Closed

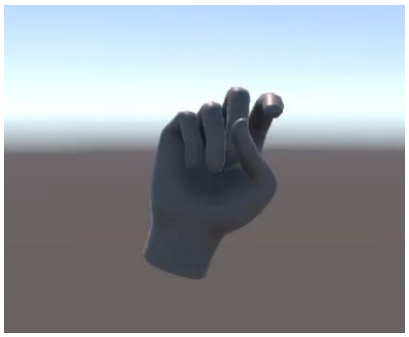
Cross subject (Training: Subject 2, Test: Subject 3, Trial 1)



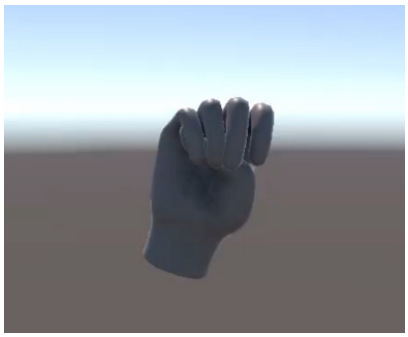
(e) Ground Truth - Pinch



(f) Prediction - Pinch



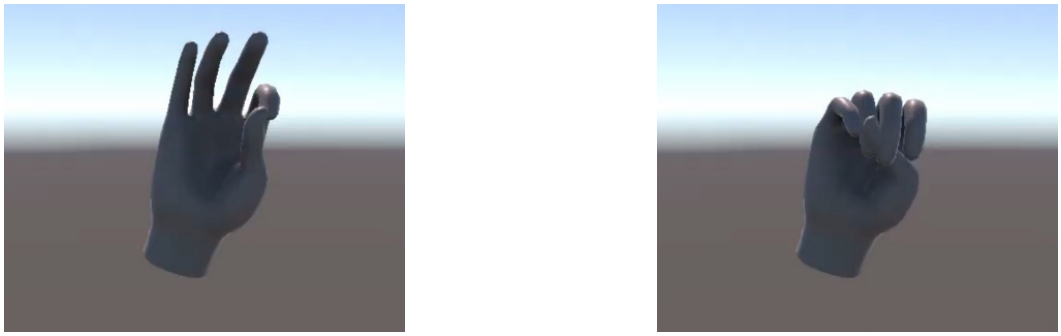
(g) Ground Truth - Hand Closed



(h) Prediction - Hand Closed

Figure 4.14 Comparison between the ground truth and the predicted hand kinematics for the concatenated motor tasks (Pinch followed by Hand Closed). The first two rows report the intra subject evaluation on Subject 2, while the last two rows show the cross subject evaluation on Subject 3 after training on Subject 2.

Multi subject Training (Subjects 2, 7, 8, 10 → Subject 3)



(a) Prediction - Pinch

(b) Prediction - Hand Closed

Figure 4.15 Predicted hand kinematics obtained using the network trained on multiple subjects (Subjects 2, 7, 8, and 10) and tested on the unseen Subject 3. The left panel shows the prediction for the Pinch phase, whereas the right panel shows the prediction for the Hand Closed phase.

to control virtual hand models in virtual reality environments.

Upon these preliminary achievements, several methodological and experimental paths should be explored in future works:

1. **Refinement of Motor Unit Decomposition and Validation**

   While the average number of identified motor units per participant was consistent with the values reported in physiological literature, the extraction process relied on some of the conditions proposed by previous researches, due to this particular sEMG dataset. Future investigations should incorporate a rigorous, manual visual inspection of the motor unit firing rates and discharge dynamics.

2. **Dataset Expansion and Synergies Correlation**

   This study is restricted to two specific hand movements: the pinch and the full hand closure. While a 2 synergy input space is sufficient to robustly distinguish and reconstruct these two tasks across different participants, this low dimensionality is a necessity forced by the lack of an inter subject alignment framework. Indeed, training the LSTM with a higher number of synergies, such as the reconstruction optimal 4 synergies identified by the R^2 metric, introduce chaotic temporal variability during cross subject validation, because the individual synergy components represent different muscle coordination processes depending on the subject.

   Future developments should focus on implementing robust synergy correlation techniques, to ensure that each input feature maintains a consistent semantic meaning across the entire subject pool.
   By expanding the kinematic dataset to include a wider variety of fine motor tasks and hand postures, a higher number of synergies can be extracted and

correlated between subjects. This expansion will allow the model to capture more complex muscle coordination patterns without overfitting to simple, low dimensional transitions.

3. **Dynamic Decomposition and Mapping**

   The current pipeline of the sEMG decomposition relies on isometric tasks, but a future goal of this research framework is to predict hand kinematics during continuous, dynamic movements. Currently, MUedit developers are integrating the work of Mendez Guerra et al. (2024) [47] to advance the decomposition algorithm in being capable of identifying active motor units in dynamic conditions. Incorporate dynamic motor unit tracking with the LSTM mapping model, proposed in this work, it will allow for a natural real time neural control of prosthetic devices and virtual avatars.

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